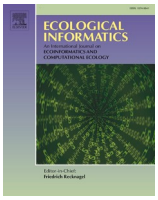
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The effect of soundscape composition on bird vocalization classification in a citizen science biodiversity monitoring project

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ABSTRACT

There is a need for monitoring biodiversity at multiple spatial and temporal scales to aid conservation efforts. Autonomous recording units (ARUs) can provide cost-effective, long-term and systematic species monitoring data for sound-producing wildlife, including birds, amphibians, insects and mammals over large areas. Modern deep learning can efficiently automate the detection of species occurrences in these sound data with high accuracy. Further, citizen science can be leveraged to scale up the deployment of ARUs and collect reference vocalizations needed for training and validating deep learning models. In this study we develop a convolutional neural network (CNN) acoustic classification pipeline for detecting 54 bird species in Sonoma County, California USA, with sound and reference vocalization data collected by citizen scientists within the *Soundscapes to Landscapes* project (www.soundscapes2landscapes.org). We trained three ImageNet-based CNN architectures (MobileNetv2, ResNet50v2, ResNet100v2), which function as a Mixture of Experts (MoE), to evaluate the usefulness of several methods to enhance model accuracy. Specifically, we: 1) quantify accuracy with fully-labeled 1-min soundscapes for an assessment of real-world conditions; 2) assess the effect on precision and recall of additional pre-training with an external sound archive (xeno-canto) prior to fine-tuning with vocalization data from our study domain; and, 3) assess how detections and errors are influenced by the presence of coincident biotic and non-biotic sounds (i.e., soundscape components). In evaluating accuracy with soundscape data (n = 37 species) across CNN probability thresholds and models, we found acoustic pre-training followed by fine-tuning improved average precision by 10.3% relative to no pre-training, although there was a small average 0.8% reduction in recall. In selecting an optimal CNN architecture for each species based on maximum F(β = 0.5), we found our MoE approach had total precision of 84.5% and average species precision of 85.1%. Our data exhibit multiple issues arising from applying citizen science and acoustic monitoring at the county scale, including deployment of ARUs with relatively low fidelity and recordings with background noise and overlapping vocalizations. In particular, human noise was significantly associated with more incorrect species detections (false positives, decreased precision), while physical interference (e.g., recorder hit by a branch) and geophony (e.g., wind) was associated with the classifier missing detections (false negatives, decreased recall). Our process surmounted these obstacles, and our final predictions allowed us to demonstrate how deep learning applied to acoustic data from low-cost ARUs paired with citizen science can provide valuable bird diversity data for monitoring and conservation efforts.

1. Introduction

Biodiversity is under profound stress from climate change, habitat loss, pollution, and other interconnected factors that now threaten approximately 25% of species with extinction and others with steep declines in abundance (Díaz et al., 2019; Dirzo et al., 2014; Leclère et al., 2020). For example, in North America, bird abundance is estimated to have decreased 29% since 1970 (Rosenberg et al., 2019) and worldwide

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there is an estimated decline in populations for 48% of extant bird species (Lees et al., 2022). To understand these changes and develop effective solutions, the science and land management communities require baseline conditions and biodiversity monitoring to track species distributions, abundances and aggregated indicators of trends at multiple spatial and temporal scales (Pereira et al., 2013). Unfortunately, adequate biodiversity data are limited, and when present, lack sustained or repeated sampling. Birds are the most widely monitored animal species (Lees et al., 2022), with a large amount of data acquired annually by a robust citizen science community, such as through the global eBird project (Sullivan et al., 2014) and the U.S. Breeding Bird Survey (Sauer et al., 2017). However, avian data collected in the field (e.g., point counts) requires an observer to accurately identify a species by sight or vocalization, with variability introduced by differences in skill level, perception of sounds, and ability to see birds (Catlin-Groves, 2012; Tulloch and Szabo, 2012). Further, data collected by volunteers tends to be spatially clustered (Mair and Ruete, 2016; Tulloch et al., 2013), has high variation in effort and observer skill (Tulloch and Szabo, 2012; Oliveira et al., 2016), and has limited temporal sampling (e.g., on weekends), thereby limiting scientific applications (Zhang, 2020).

Sound autonomous recording units (ARUs) placed in the field for days to months can provide long-term and systematic data on the occurrence and abundance of sound-producing wildlife, including birds, amphibians, insects and mammals (Balantic and Donovan, 2020; Gibb et al., 2019). Further, ARUs can provide similar estimates of bird richness as traditional point counts (Darras et al., 2018) and sound recordings are a permanent record that can be re-interpreted as necessary (Aide et al., 2013; Gibb et al., 2019). Recorders can be placed on public lands by volunteers, but also delivered to private landowners for deployment, thereby increasing sampling across a study region (Snyder et al., 2022). Leaving ARUs operative for days can generate an immense amount of data, which becomes a challenge to manual identification of species by their unique vocalizations, necessitating automated machine-based detection techniques (Brandes, 2008). Although human identification of species in sounds has the advantage of much higher flexibility in the interpretation of sounds, machine-based detection could potentially overcome the challenges of bias due to variance in observer skills. For example, in a study of bat sounds, Jennings et al. (2008) found machine-based identifications overperformed humans in accuracy of detections. Further, despite the differences in detection probability between traditional bird point counts and ARU surveys, presumably due to birds altering their behavior in the presence of humans, Furnas and Callas (2015) found no differences in species occupancy estimates for a limited number of species. Thus, the advantages of using ARUs are mainly the permanent record of the sample, reduced bias (spatial, temporal, observer skill), and cost-effectiveness.

Automating species detections in sound recordings is a growing field, and encompasses a range of taxonomic groups, including birds (Acevedo et al., 2009; Campos-Cerqueira et al., 2017; Furnas and Callas, 2015), bats (Walters et al., 2012), amphibians (Acevedo et al., 2009; Campos-Cerqueira and Aide, 2017), and insects (Aide et al., 2017). Priyadarshani et al. (2018) reviewed first generation techniques, which include steps to reduce noise, segment bird vocalizations from background, and extract pre-determined features for training machine learning classifiers. These features tend to be manually identified by domain experts, limiting performance compared to more recent approaches that learn the features (Gupta et al., 2021). Further, much of this early research used relatively small datasets with low levels of noise and involved manual tuning of parameters. As a consequence, the techniques generally have poor recall rates (i.e., ability to detect all bird sounds), high sensitivity to noise, and do not generalize with new species, noise conditions or recording equipment (Priyadarshani et al., 2018; Stowell et al., 2019).

Modern deep learning classification based on convolutional neural networks (CNNs) is relatively new to the field of bioacoustics species detection (see review by Stowell, 2022), and tends to have very high (>90%) accuracy for bird species (Incze et al., 2018; Kahl et al., 2021; LeBien et al., 2020; Ruff et al., 2021; Stowell et al., 2019; Zhong et al., 2020). Tuning of the network hyperparameters and convolutional filters provides automated feature learning, thereby removing an obstacle of earlier machine learning methods (Gupta et al., 2021). Sound data include frequency, amplitude and temporal information, and there are different approaches to express these data in CNN architectures. One approach is to represent sound data as amplitude-time waveforms, then train 1-dimensional neural network architectures, such as Recurrent Neural Networks (RNNs) to identify distinct patterns (Sang et al., 2018; Weninger and Schuller, 2011; Zhang et al., 2021). Another approach is to train CNNs to classify 2D image spectrograms (Kahl et al., 2021; LeBien et al., 2020; Ruff et al., 2021; Stowell et al., 2019; Zhong et al., 2020). A popular extension is to use mel-spectrograms from human speech and music domains, which in the process of computing the spectrogram, recorded sound frequencies (Hertz) are remapped to a frequency scale (mel scale) whose intervals are perceptually equidistant to human hearing (Gupta et al., 2021; Kahl et al., 2021; Lasseck, 2019; LeBien et al., 2020; Zhong et al., 2020). The use of 2D CNN models has dominated the annual LifeCLEF Bird Recognition Challenge (BirdCLEF) competition, with an important finding being that architectures adapted from the image recognition domain do not necessarily need to be deep in order to effectively classify calls and songs for all bird species (Kahl et al., 2021), which could simplify model training and implementation.

In our review of peer-reviewed research, we found that a common deficiency with acoustic bird classification research is the reporting of overly-optimistic accuracy or a focus on the accuracy metrics that may not be ideal for biodiversity applications. A common approach is to use a hold-out set randomly sampled from the same dataset used to train the models, as a single set (e.g., Zhong et al., 2020) or in a k-fold sampling scheme (e.g., Gupta et al., 2021). These test clips are often clean examples of bird calls and exclude those with non-target sounds, overlapping sounds, and background recorder noise, and thus do not depict actual accuracy when a classifier is deployed on whole recordings in a real-world application. There has been some progress on reporting more realistic accuracy in applied settings. LeBien et al. (2020) assigned class presence within a larger temporal window (i.e., 1 min) based on thresholds on class probability from finer temporal predictions, i.e., 2 s with 1-s overlap. However, this method reports accuracy with a different resolution than the classifier and forces a constraining temporal aggregation. In a CNN-based study focused on bird and other small mammal vocalizations, Ruff et al. (2021) created a large independent dataset of spectrogram segments through manual validation, half of which were non-target sounds and background noise, and 6% included multiple target species. The applied CNN had 12-s predictions but did not match the variable temporal sampling of the test data. Since 2019, the BirdCLEF competitions have evaluated accuracy with independent "soundscape" test data, which are continuous recordings with detailed species annotations from expert ornithologists (Kahl et al., 2020). These test data are thus a domain shift, and provide an assessment of accuracy when the classifier is applied to recordings from new environments from which it was trained (Kahl et al., 2021). Further, it is important for accuracy statistics to be relevant to a particular biodiversity application (Knight et al., 2020a). For many biodiversity applications, such as occupancy modeling or species distribution models, there is a general requirement to correctly identify the presence of a species from acoustic data (Campos-Cerqueira and Aide, 2016), but not necessarily all instances of a species vocalizing in a recording. High detection accuracy reported in some studies come at the cost of high numbers of false positives that can lead to very inaccurate estimates of occupancy and biodiversity (Royle and Link, 2006).

Many 2D CNNs use well-labeled training and validation data from the study domain intended to optimize weights in existing image-based architectures (e.g., ResNet, VGGish, MobileNet), i.e., transfer learning. These data are usually hard to acquire and thus have low sample sizes. This often leads to the use of larger labeled data from outside the study domain. Recent BirdCLEF competitions use weakly-labeled audio

recordings from the open-access xeno-canto (XC) archive (www.xeno-canto.org) for bird species as well as other public audio archives for background sound, including other animals (Kahl et al., 2020). These data span a wide range of environmental conditions and recorder types, allowing the training of a generalized bird species classifier (Henkel et al., 2021; Kahl et al., 2020; Kahl et al., 2022; Lasseck, 2019). Although BirdCLEF accuracy is now tested with independent soundscapes, the accuracy of predictions can increase by over 25% when some training data come from the soundscapes themselves (Lasseck, 2019). Large datasets can also be used in an initial pre-training step applied to initial pre-trained CNN backbones, which can help learn low-level spectrogram features, followed by fine-tuning with data from target classes from within the study domain (Yosinski et al., 2014; Quinn et al., 2022).

Here we report on a 2D CNN acoustic classification pipeline for detecting bird species in Sonoma County, California, USA. The sound and reference data were collected with the help of citizen scientists within the *Soundscapes to Landscapes* project (www.soundscapes2landscapes.org). This aspect of our research introduces several challenges not found in other studies, such as the use of low-cost recorders, placement of recorders in various public and private landscapes along an urban-wildland gradient, and reference vocalizations collected with varying levels of expertise. An advantage of our study is a collaboration with expert citizen scientists to label every bird vocalization in a sample of 1-min recordings (i.e., soundscape data), providing independent test data for assessing real-world accuracy of our classifiers in our study domain. We used weakly labeled data from outside our study domain to pre-train three image-based CNN classifiers to recognize spectrogram-based features, followed by fine-tuning with data from our study domain. In contrast, other studies do not include an intermediary pre-training step prior to fine tuning. Further, we take advantage of a companion study by Quinn et al. (2022) that used a deep learning classifier to detect general soundscape composition components in our audio data. We use these data to investigate potential sources of coincident sounds that may cause errors in our bird species classifier. To our knowledge, the pairing of bird species and soundscape deep-learning classifiers in a unified analytical framework is a unique contribution of our study.

We trained three ImageNet CNN architectures (MobileNetv2, ResNet50v2, ResNet100v2), which function as a Mixture of Experts (MoE), for classifying bird species in our study area. The primary objectives of our study are to: 1) assess classifier accuracy with fully-labeled soundscape recordings, and compare this accuracy to an estimate from a reserved set of reference data that were also used to train CNNs, a common practice in this domain; 2) evaluate the potential improvement in accuracy by pre-training CNN models using bird vocalizations from weakly-labeled XC audio recordings; and, 3) assess how detections and errors are influenced by the presence of coincident biotic and non-biotic sounds (i.e., soundscape components). To have a frame of reference on the relative precision and improvements on our models, we also compare the performance of our approach against an external standard. For this we used BirdNET, a global acoustic bird 2D CNN that is also being developed for citizen science sound data (Kahl et al., 2021). It is not our intent to evaluate BirdNET performance, as it was not trained with our data.

2. Materials and methods

2.1. Field data collection

In situ acoustics data were acquired in Sonoma County, in northern California, USA. Sonoma County is 4118 km^2 with a wide variety of land use/cover types, including agricultural, urban - residential, forest, chaparral and grassland habitat types (Fig. 1). Recordings were collected by citizen scientists from 2017 to 2021 during the breeding bird season (mid-March thru mid-July) when birds tend to vocalize frequently. To collect acoustic field data, we used two types of ARUs: Android-based LG smartphones with attached microphones and waterproof cases and

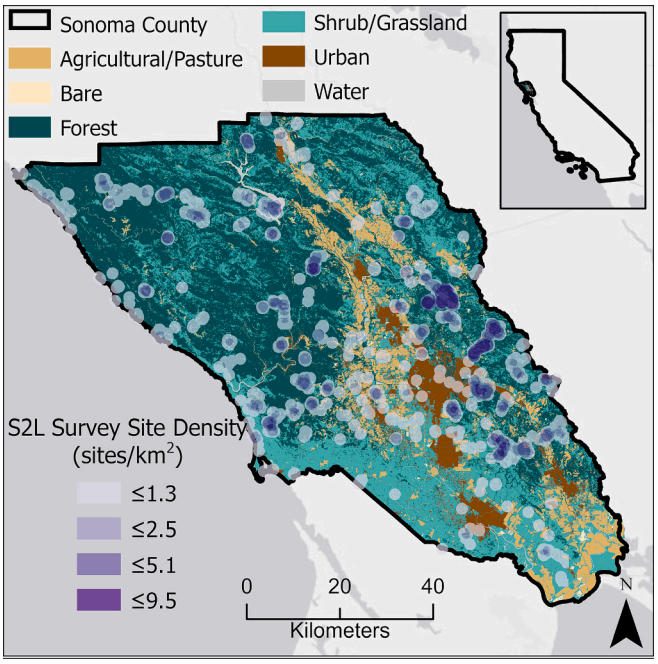


Fig. 1. Study area and sound recorder sampling density (number of sites per km^2) in Sonoma County (n = 1236). Land cover layer from Sonoma County Agricultural Preservation and Open Space District.

AudioMoths (Hill et al., 2019) enclosed in nylon pouches. ARUs were deployed at sites for 3 to 5 days, and programmed to record 1 min of every 10, thus providing temporal sampling through day and night, with an average of 582 min of recordings per site. There was a total of 735,984 min of recordings from 1236 sites (Fig. 1), with 81% from AudioMoth and 19% from LG ARUs. Additional details on field data collection can be found in Snyder et al. (2022).

2.2. Reference bird vocalizations

2.2.1. Regions of interest (ROI) for classifier fine-tuning training and testing

We targeted 54 species that have distinctly unique vocalizations (call or song pattern) and are common in the study area during the breeding bird season (Table 1). Although additional species could have been studied (~180 species total; Sonoma County Breeding Bird Atlas, 2020), our project did not attempt to be exhaustive given the constraints of a citizen science project based on volunteer labor. Some species include 2 to 3 different vocalizations (Table S1). Reference bird vocalizations were collected by citizen scientists using a web-based interface in the Arbimon platform (Aide et al., 2013). In this process, an expert citizen scientist with knowledge of local bird vocalizations identified parts of spectrograms that represent a target bird vocalization. The system then used this template region of interest (ROI) and a pattern matching algorithm (LeBien et al., 2020) to find similar ROIs in our archive of sound recordings. Citizen scientists then validated whether the matched ROIs contained the target vocalization by visual observation of the spectrogram and/or listening to the sound clip. We used a consensus-based vote for validations from users without bird knowledge, taking the majority of present or absent votes for up to 5 votes. Expert votes were taken as truth (i.e., no consensus). More details on this process and final ROIs are available in Supplemental Section 1 and in Snyder et al. (2022). Importantly, by using pattern-matching we might have introduced a sampling bias because it finds ROIs that are relatively clean due to the use of a correlation threshold. Despite data augmentation methods, this bias could lead to potentially lower precision in our models when tested against a random set of recordings.

Table 1
Summary of soundscape data, including species, species codes, total number of detections, percent of labels in the dawn chorus (6 a.m. to 9 a.m.), and whether the species was evaluated in our soundscape analysis. Four-letter codes follow Pyle and DeSante, 2003.

Species	Species code	Total detections	Percent dawn chorus	Training call detections	Soundscape analysis
Acorn Woodpecker	ACWO	642	57	631	√
American Crow	AMCR	459	58	392	√
American Robin	AMRO	965	82	859	√
Anna's Hummingbird	ANHU	64	70	0	–
Ash-throated Flycatcher	ATFL	7	14	7	–
Bewick's Wren	BEWR	1414	80	1098	√
Blue-Gray Gnatcatcher	BGGN	73	79	64	√
Black-headed Grosbeak	BHGR	693	64	599	√
Brown Creeper	BRCR	447	65	23	–
Band-tailed Pigeon	BTPI	27	56	27	–
Black-throated Gray Warbler	BTYW	95	82	62	√
Bullock's Oriole	BUOR	21	14	3	–
California Towhee	CALT	977	66	630	√
California Quail	CAQU	183	89	94	√
California Scrub-Jay	CASJ	305	78	235	√
Cassin's Vireo	CAVI	594	86	515	√
Chestnut-backed Chickadee	CBCH	1064	74	860	√
Common Raven	CORA	362	54	273	√
Common Yellowthroat	COYE	80	64	67	√
Dark-eyed Junco	DEJU	933	71	674	√
Eurasian Collared-Dove	EUCD	159	69	72	√
Great Horned Owl	GHOW	6	17	0	–
Grasshopper Sparrow	GRSP	20	60	0	–
Hermit Thrush	HETH	168	84	0	–
House Finch	HOFI	496	61	230	√
House Wren	HOWR	197	70	0	–
Hutton's Vireo	HUVI	306	40	0	–
Lazuli Bunting	LAZB	32	91	22	–
Marsh Wren	MAWR	148	45	130	√
Mourning Dove	MODO	106	59	93	√
Mountain Quail	MOUQ	53	51	40	√
Northern Flicker	NOFL	116	47	31	–
Nuttall's Woodpecker	NUWO	31	61	24	–
Oak Titmouse	OATI	817	53	565	√
Orange-crowned Warbler	OCWA	709	67	428	√
Olive-sided Flycatcher	OSFL	1	0	0	–
Pacific Wren	PAWR	70	47	54	√
Pileated Woodpecker	PIWO	9	67	4	–
Pacific-slope Flycatcher	PSFL	1335	82	726	√
Red-shouldered Hawk	RSHA	20	100	20	–
Red-winged Blackbird	RWBL	1284	44	172	√
Savannah Sparrow	SAVS	57	51	41	√
Song Sparrow	SOSP	232	54	170	–
Spotted Towhee	SPTO	673	64	575	√
Steller's Jay	STJA	693	66	371	√
Swainson's Thrush	SWTH	93	84	15	–
Warbling Vireo	WAVI	459	57	459	√
White-breasted Nuthatch	WBNU	236	38	226	√
White-crowned Sparrow	WCSP	160	72	93	√
Western Meadowlark	WEME	368	38	322	√
Western Tanager	WETA	131	87	126	√
Wild Turkey	WITU	54	96	54	√
Wilson's Warbler	WIWA	228	46	185	√
Wrentit	WREN	120	68	102	√
Totals		18,992		12,463	37 species (√)

2.2.2. *Soundscapes for testing the classification in field conditions*

To generate independent data to test the accuracy of the CNN model predictions in field conditions within our study area, citizen scientist experts created "soundscape data", which was an exhaustive review of a stratified random sample of 661 recordings (of 1-min) with every bird vocalization labeled by the second when it started. More information about the sampling scheme used to select recordings is provided in Supplemental Section 2. We compiled 18,992 detections of our 54 species in the soundscape data (Table 1). Of these, 37 species had at least 40 vocalization events in the soundscape data for the same call/song types used in training, and thus we limited our soundscape analyses to these species, with data from 628 recordings. Because we collected ROI and soundscape data simultaneously, there were 206 recordings in common for ROI training and soundscape data. This overlap was limited to 473 out of 32,704 ROIs (1.5%) for the 37 target species in our soundscape analysis, or 473/44,261 (1.1%) of all ROI training data, available prior to data augmentation. We acknowledge this artifact could lead to higher estimates of model precision but note that 1) soundscapes were labeled independently from ROI labels, 2) data augmentation diluted bias from these 473 training ROIs, and 3) we found that there was a large difference in soundscape and ROI accuracy, thus revealing that the effect of the overlap was minimal. Further, since these soundscape data are from the same audio domain as the ROI data, accuracy statistics with these data do not indicate the transferability of our CNN processing pipeline outside our study domain (e.g., a different location, time period, or recording device), which differs from the "domain-shift" soundscape data used in Kahl et al. (2021) and BirdCLEF competitions (Kahl et al., 2020).

2.3. Mel spectrogram 2D representations of sound data

All training and inference in this study was performed using 2-s segments of our 1-min recordings. This segment length was determined by the relatively small widths of our pattern-matching templates (mean width 1.3 ± 1.0 s across species), initial empirical experimentation, and a consideration of methods by LeBien et al. (2020), which used the same pattern-matching routine for reference data collection.

Our field recordings were acquired at sample rates of either 48 kHz (AudioMoth) or 44.1 kHz (LG smartphone). Vocalizations of our 54 species did not contain frequency components above ~11 kHz and so we resampled all our audio data to 22.05 kHz (twice the sampling rate needed to represent our vocalizations). Mel spectrograms were computed for 2-s audio clips using the Python librosa package (see supplemental code for parameters). We used the square of the mel spectrogram energy values in dB (i.e., the power of the signal) as this results in visually more distinctive bird calls. Since our CNNs operated on color images, having been pre-trained on the ImageNet dataset, we converted the 2-s mel spectrograms to 3-channel (RGB) images using librosa's default magma colormap, which is perceptually uniform (Fig. 2; Fig. S1).

2.4. CNN pre-training data from xeno-canto recordings

For pre-training of our CNN classification models, described in Section 2.6, we used recordings of bird vocalizations in the XC archive. These recordings were gathered with various devices and quality, but required 100% positive identification of bird species. We used a curated dataset of XC recordings from the 2020 Cornell Birdcall Identification competition (Howard et al., 2020), from which we selected recordings for 40 species that were targeted in our study and found in the archive. These XC recordings were a domain shift from our recordings, as we focused solely on Sonoma County and used only two types of recording devices. Unlike our ROI data, the XC recordings were long segments and not focused on a specific part of a bird's vocalization and thus we sought an automatic means of isolating discrete sounds within variable-length recordings. This was accomplished with the *auto_detec* function in the R warbleR package (Araya-Salas and Smith-Vidaurre, 2017), which finds the start and end of vocalizations based on amplitude, duration, and frequency range parameters. We analyzed signals with an amplitude threshold of 5%, duration of signal detection (i.e., vocalization duration) from 0.1 to 2 s, and frequency bandpass of 2 to 9 kHz. The *auto_detec* function yielded 444,846 isolated signals, or start/stop times of vocalizations.

For generating mel-spectrograms from XC vocalizations, we padded each side of vocalization to be at least 2 s in length, clipped the recording from the original .wav file, and then appended these clips into one recording per species. Mel spectrograms were generated at a 2-s interval from this combined recording following the parameters described in Section 2.3. We randomly sampled XC data by species to create a training and validation set. If there were >750 vocalizations for a species, 150 vocalizations were randomly sampled for a validation set; if there were fewer than 750 vocalizations, we used 20% of the total for validation. The remaining XC data were used for training, with an average of 1608 (minimum = 33; maximum = 5540) per species. We did not need independent XC test data as accuracy improvements with pre-training were evaluated with ROI test and soundscape data (Section 2.8).

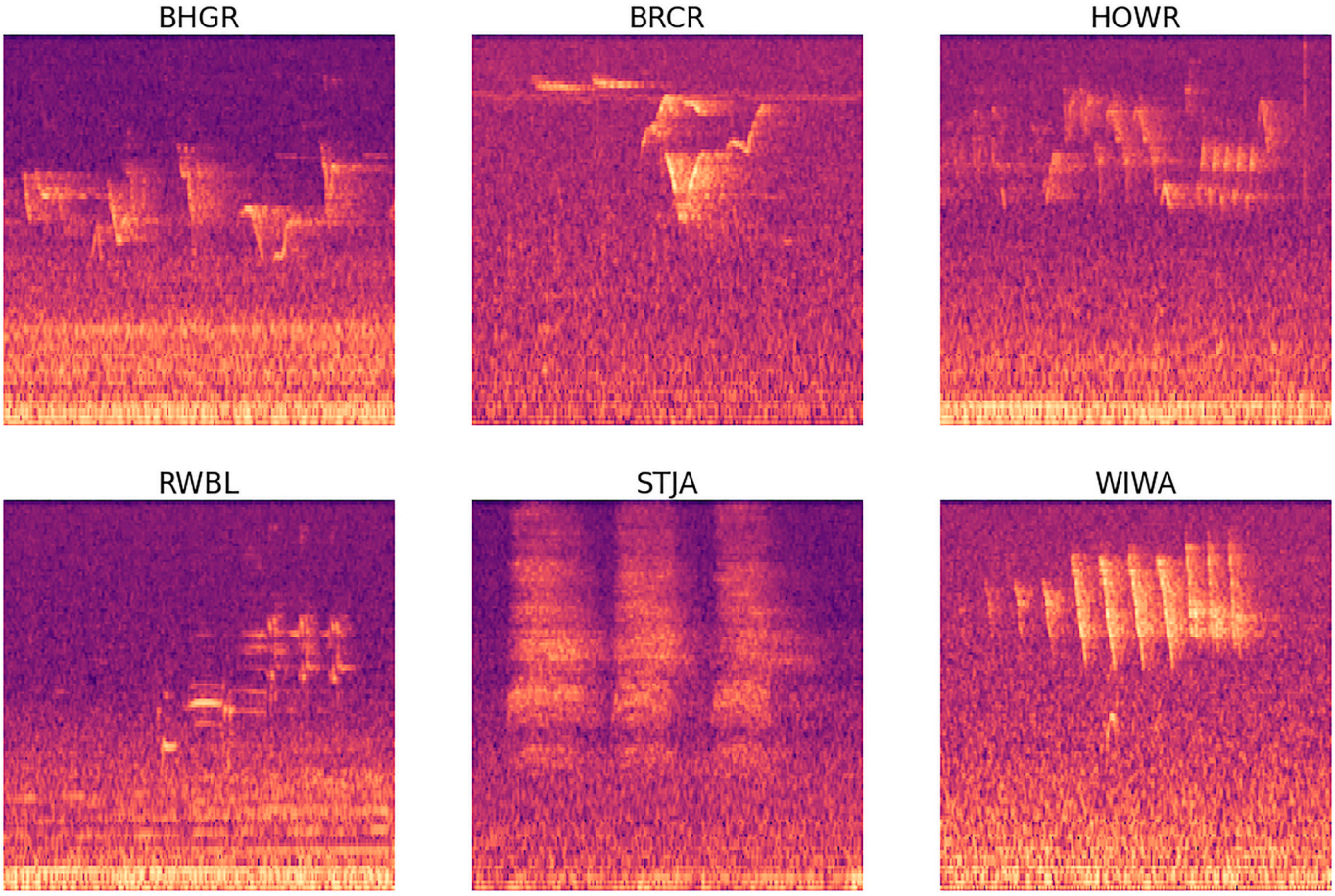


Fig. 2. Example 2-s mel-spectrograms of bird vocalizations for six species in our study.

2.5. CNN fine-tuning data from ROIs

We randomly sampled 50 ROIs per species with "presence" to form a held-out test set for accuracy assessment (Section 2.8.1). The remaining ROIs with species presence were used to fine-tune the CNN models. We randomly sampled 100 of these ROIs per species to form a validation set and the remaining formed a training set, which had on average 887 per species (min = 349, max = 3094). ROIs that were >2 s in duration were used in their entirety. For those that were <2 s in duration, we expanded the start/stop (i.e., padded) equally on each side of the ROI to total 2 s. Different vocalizations for a given species were combined into a single class per species. If any ROIs were overlapping in time from different pattern-matching runs for a given species, we placed those ROIs in the training set to avoid temporal autocorrelation in our validation and test data.

The recordings for each ROI were cut using start/stop times and sequentially appended to create a single merged .wav file for each species and set (i.e., train, validation) combination. Each merged .wav file was then sliced into 2-s increments and mel-spectrograms computed. Note that since there were some ROIs from the pattern matching that were >2 s, some 2-s recording clips included parts from different ROIs.

Performing data augmentation on training datasets is a widely used technique in deep learning, including sound classification applications, to regularize models and provide more robust features (Salamon and Bello, 2017). We performed data augmentation by splitting the merged .wav training file for each species in half and then adding these two halves sample-wise. This augmented .wav file was similarly split into 2-s increments as above and mel-spectrograms were computed. The training dataset increased by 50% after this data augmentation, which is important for small training datasets such as ours.

At the end of this process, we had training and validation, 2-s mel-spectrograms that were used to fine-tune the CNN models.

2.6. CNN multi-species classification models

We trained three 2D, image-based CNN models to classify bird species based on their vocalizations represented in mel-spectrograms. These were multi-species models, although we ultimately chose the best model for each species using soundscape data (Section 2.8.3). The underlying network backbones were pre-trained on the large ImageNet archive of labeled images (1.4 M images across 1000 classes) and included MobileNetv2 (Sandler et al., 2018) with 4.44 M parameters, ResNet50v2 (He et al., 2016) with 23.65 M parameters, and ResNet101v2 with 42.71 M parameters. We chose these architectures as they were widely used in the literature (Stowell, 2022) and provided a contrast between smaller (more generalist) and larger (more specialist) architectures. When there is a large amount of training data for transfer learning, the larger ResNet networks perform better than the smaller MobileNet networks due to their capacity (Khan et al., 2020). However, in our case, the ROI reference data available for training were limited, so we sought to explore different network sizes in conjunction with pre-training with additional acoustic reference data from outside of our audio domain, followed by fine-tuning with ROI data from our study site.

We evaluated model performance with and without additional pre-training with external sound data prior to fine-tuning with sound data from our study. For the pre-trained models, we used training and validation XC mel-spectrograms to adjust the weights of each of the three CNN models. Pre-training with XC data was implemented for all CNN models starting with training the final, dense layer for 10 epochs followed by training all layers for 10 additional epochs.

Fine-tuning using the ROI training and validation mel-spectrograms was then applied to both the default CNN backbones (i.e., not pre-trained on XC) and to the XC pre-trained CNNs. The fine-tuning process consisted of two stages: (1) model training on the classifier only (i.e., dense, fully-connected layer), freezing all other trainable parameters in CNN layers for 10 epochs (learning rate = 0.0001), and (2) fine-tuning the network by unfreezing all layers and training for 10 more epochs. Model weights were saved only when the internal validation loss decreased.

All fully-trained CNNs produced a vector of 54 predicted probabilities using a binary cross entropy classifier with sigmoid activation (one value for each species). We applied thresholds to these probabilities to evaluate model accuracy (Section 2.8).

2.7. Inference on whole recording archive

We applied the fine-tuned CNNs at a 2-s window with 1-s overlap, yielding 59 mel-spectrograms for each 1-min recording. This resulted in >43 million mel-spectrograms from all recordings. Inference on each mel-spectrogram generated 54 probability values (one for each species), for each of the three models, with fine-tuning with ROI data on both default ImageNet and XC pre-trained versions. Given this large data volume to process and analyze, we performed CNN inference on a high-performance cluster and stored our results in a MySQL database for rapid retrieval. We further pre-filtered all predictions into an informative set by archiving only those with a sigmoid probability ≥0.65.

2.8. Evaluation of accuracy

2.8.1. Evaluation of accuracy with ROI test data

The ROI test data consisted of 50 presences and 50 absences for each species. In the accuracy assessment, we used the ROI length as determined by pattern matching template for a given bird species. Presences were the group of randomly-sampled ROIs set aside for testing that were never used to train the models. We assumed that if the species was evaluated by citizen scientists as absent in the ROI, then the target bird was not there. Two possible sources of error might have been introduced when determining absences for a target species. First, our citizen scientists could have failed to detect the target species in the ROI due to noise or otherwise unclear patterns. Absences were based on consensus votes or expert review, and we are confident that this source of error is minimal. Second, our citizen scientists assessed matches for a species within the range of frequencies found in its pattern matching template, which was focused on common vocalizations in our region. Birds, among other animals, are known to shift frequencies in order to adapt to a noisy or competitive soundscape (Bermúdez-Cuamatzin et al., 2011; Roca et al., 2016). It is possible that the target birds were not detected by pattern matching if their vocalizations were outside the range of the template, and therefore never assessed by the citizen scientists. However, the range of frequency shifts reported in the literature are relatively small (200–500 Hz, e.g. Luther et al., 2016) compared to the frequency ranges in our templates (~1.5 to 7 KHz, depending on the species). Any frequency shift would likely be included in the template frequency range and inspected by citizen scientists as a presence or absence. Further, because of the county scale of our sound collection, we do not expect that variations in vocalization (e.g., dialects) for a given bird species would fall outside of the search frequency-band for a given species. Thus, we are confident that a detection of a target bird for an absence is indeed a FP.

We randomly selected absences from the same set of recordings in which we found presences across all species. A given recording may or may not have included absences and presences for the same species, and may contain labels from one or more species. In the accuracy assessment, the species prediction for an ROI was retrieved from a CNN model for species detected with probabilities ≥0.65 (i.e., 65%). That is, for each ROI of a species, we took the highest value of the predictions for that species at the center second and one second before and after the center of the ROI. We also retrieved the predicted probabilities for all other species in the same time interval. The 0.65 minimum threshold was intended as a filter for informative rather than random (i.e., ≤0.50) predictions. We calculated true positives (TP), false positives (FP), false negatives (FN), precision (TP/[TP + FP]), and recall (TP/[TP + FN]) for

a range of CNN probability thresholds (i.e., 0.65 to 0.99). For ROI presences, if a species was detected with a CNN probability higher than a given threshold that matched the target ROI species label, the sample was a TP; otherwise, the sample was considered a FN for that species. For ROI absences, if a species was detected that matched the target ROI label, the sample was considered a FP for that species. We did not expect the ROI data to adequately represent species vocalizations that overlap those from other bird species, or soundscape components more broadly. Instead, our ROI test data included only relatively clean (as perceived by a human observer) presences and absences. It follows that the performance results against the ROI data would be higher than the more realistic soundscape data, which were from an exhaustive review of recordings for target vocalizations and include samples with a range of soundscape conditions (Section 2.8.2).

2.8.2. Evaluation of accuracy with soundscape data and BirdNET

A benefit of our soundscape data was that they included labels of all bird vocalizations (i.e., calls and songs) within a 1-min recording, including those with temporal overlap. We matched manually-detected soundscape vocalizations to 2-s predictions found within a 1-s and 3-s window before and after the vocalization's start time, respectively. The species label of the soundscape vocalization and all probability values within the temporal window were used to determine the accuracy of the model predictions over a range of probability threshold values, as we did with the ROI test data. These predictions were used to evaluate total precision and average species (class) precision. Although soundscape data used in our accuracy assessment included vocalizations that overlap in the same time period of the recording, we did not assess this factor in our analyses.

In comparing ROI and soundscape accuracy, we acknowledge that some difference could be related to variable sample size among species between the two datasets. However, we believe this effect is minimal as we used well-randomized representative samples of soundscapes and report total precision and recall across species. Since these measures are aggregated raw counts of TP, FP, and FN, if our soundscape data and ROI test datasets were properly randomized and if they result in representative performances for each class, it follows that the difference in performances shown in the evaluation metric curves is expected to be insensitive to class imbalances in the soundscape data. Further, we report average class precision and recall statistics at optimal thresholds (Section 2.8.3), which are less influenced by differences in positives in each dataset.

We also compared the soundscape accuracy of our XC pre-trained, fine-tuned CNN models (hereafter CNN-XC) against the leading BirdNET CNN classifier (Kahl et al., 2021). This is a ResNet, spectrogram-based multi-species classifier of 984 birds from North America and Europe. We used the BirdNet Analyzer version 2.2 (https://github.com/kahst/BirdNET-Analyzer) to predict bird species in our soundscape data. The BirdNET processing pipeline is designed to make predictions in 3-s clips, which we processed with 2-s overlap, resulting in 57 samples for each soundscape recording. We ran the process with default settings with and without geographic location set for the center of Sonoma County (38.5780° N, 122.9888° W), and considered all species predicted over a given sigmoid probability threshold within each sample as present. We used start times of soundscape vocalizations for matching BirdNET species predictions. To better match our process of evaluating soundscape accuracy with our CNN models, we split each BirdNET 3-s prediction into two 2-s predictions with 1-s overlap. Precision and recall curves were generated by varying the BirdNET probabilities as we did with our CNN models, for both predictions with and without the geographic filter. In contrast to the evaluation of our CNN models, when testing BirdNET predictions we included all vocalization types for our target bird species (i.e., we did not filter them to those used in training our CNN models).

2.8.3. Optimal model and threshold selection

Our next step was to evaluate the effect of soundscape components on the precision and recall of our CNN-XC models (Section 2.9). To do this, we sought an optimal model and threshold for each species separately. Most studies seek models that balance precision, or how many positive predictions are correct, and recall, or how many of the true predictions are retrieved (also referred to as sensitivity or true positive rate). In this context, accuracy is often reported as a Fβ score, or $(1+\beta^2) \times (\text{precision} \times \text{recall})/(\beta^2 \times \text{precision} + \text{recall})$, which gives equal weight to both measures when $\beta = 1$, i.e., the F1 score. Precision is a more important metric for most applied biodiversity applications as it decreases with more FP, i.e., the species being predicted as present when it is not there. A more practical target for biodiversity monitoring could be stated as, for example, a classifier with >80% precision and > 20% recall, or an alternative form of Fβ score, such as β=0.5 (F0.5), which places more weight on precision over recall (Kahl et al., 2021). Here we report optimal probability thresholds by selecting those that maximize F0.5 scores using soundscape data. Precision and recall were computed for each CNN model on a species (class) level, as well as for all species combined (total precision and recall).

For the 37 species in our soundscape analysis (Section 2.2.2), we picked the model and associated threshold that produced the highest F0.5 score. For the remaining 17 species, we averaged maximum-F0.5 probability thresholds from MobileNet, since this architecture was selected as optimal for the majority of species. Future research could sample more soundscapes to retrieve sufficient detections needed to refine optimal model selection and thresholds for these 17 species. We applied these optimal models and thresholds to soundscape ($n = 37$ species) and ROI data ($n = 54$ species) and reported class precision and recall.

2.9. Analysis of bird detections with coincident soundscape components

We sought to provide additional insight on how bird detections from optimal CNN models were influenced by other types of sounds. For this analysis, we associated bird species detections and non-detections from our soundscape data with the presence of soundscape components: human noise from automobiles, airplanes, etc. (Anthropophony); wildlife vocalizations, including birds, amphibians and insects (Biophony); weather phenomena, such as wind and rain (Geophony), quiet periods with mainly background recorder noise or faint sounds (Quiet); and, abrupt noise from physical contact, such as from branches banging the recorder from wind (Interference). These soundscape components are collectively called ABGQI. Our companion study by Quinn et al. (2022) describes using an independent CNN and associated thresholds developed for the same sound data in our study domain. Briefly, the ABGQI classier was pre-trained on XC data and fine-tuned with sound reference data from our audio data (same domain). The classifier achieved an overall F0.75-score of 0.88, precision of 0.94, and recall of 0.80 across the ABGQI classes.

The ABGQI classifier was then applied to the same 2-s clips used for bird detections. We used the search window for matching bird detections to our soundscape data (Section 2.8.2) and marked if a given sample had any ABGQI classes detected in the search window. Soundscape components were associated with labels of TP, FP, or FN for bird species within samples, allowing us to look at how bird detections and errors are associated with coincident biotic and non-biotic sounds. We tested the influence of ABGQI and species on TP, FP and FN, respectively, using separate logistic generalized linear mixed models for each case (see Supplemental Section 3). The "TP model" set as response value 1 for TP and FN or FP were assigned 0. This inference sought to understand the effect of soundscape components in obtaining a correct classification vs not. The "FN model" treated FN as 1 and TP as 0, and did not include FP data, so that inference was limited to understanding the effect of ABGQI on a resulting FN instead of a correct detection. Similarly, the "FP model" treated FP as 1 and TP as 0 to evaluate the effect of

sound components on error of commission vs a correct detection and did not include the FN data. In all models, species and 1-min recording name were treated as random effects. Note that recording name encompasses the variation among sites, temporal differences, and recorder type (LG vs. AudioMoth), so it accounts for more than just possible error correlations from the same recording. The random species effect accounts for any species-level effects on the results.

3. Results

3.1. Effect of pre-training CNN models with xeno-canto data prior to fine-tuning

When assessed with test ROIs, the pre-training of CNN models with auxiliary XC mel-spectrograms, followed by fine-tuning (i.e., CNN-XC models) with training ROIs improved total recall (all species together) by an average of 5.9% and slightly increased total precision by an average of 0.2% relative to not including the XC pre-training step, considering all three models and probability thresholds from 0.65 to 0.99 (Fig. 3 - ROI). However, when tested against the soundscape data, there was an average 0.8% decrease in total recall and an average 10.3% increase in total precision with XC pre-training (Fig. 3 - Soundscape). Given that we sought high precision when applied to a real-world inference scenario (i.e., a soundscape), we focus our results on CNN-XC models for the remainder of our analysis.

3.2. Differences between ROI and soundscape accuracy with CNN-XC models

When assessed for all species, we found total precision (all species combined) as measured using ROI test data was very high across all probability thresholds (Fig. 4 - ROI). For example, at a threshold of 0.99, total precision and average species precision with ROI data was 99.4% and 97.3% for MobileNet, 99.2% and 99.1% for ResNet50v2, and 99.1% and 99.1% for ResNet101v2, respectively. In contrast, when tested using all soundscape data, precision was considerably lower (Fig. 4 - Soundscape). At a 0.99 threshold, total precision and average species precision with soundscape data was 76.7% and 86.6% for MobileNet, 78.5% and 88.8% for ResNet50v2, and 76.1% and 86.6% for ResNet101v2, respectively. In the worst case, species PAWR at the 0.99 threshold had an average precision of 100.0% across models when tested with ROI data, while average precision was 48.7% with soundscape data. We thus conclude that the standard approach to estimating accuracy with a hold-out set of sample data from the same dataset as the training/validation data (i.e., ROIs) greatly overestimates performance of CNN models.

3.3. Comparison with BirdNET

When tested against soundscape data, all three of our CNN-XC models had slightly lower total precision than BirdNET with the geographic filter (max = 81.8% at 0.65 threshold), and CNN-XC models performed better than BirdNET without the filter (max = 69.6% at 0.74 threshold; Fig. 5A). However, all CNN-XC models had higher recall than both BirdNET versions, regardless of thresholds. At a 0.99 threshold, CNN-XC models had an average 13.9% higher total precision and 27.6% higher total recall than BirdNET with the geographic filter (Fig. 5B). We found BirdNET total precision had very little sensitivity to thresholds up to 0.90, while precision declined at higher thresholds as FN increased more than FP (Fig. 5A, BirdNET). The BirdNET classifier with a geographic filter had sufficient total recall, which we set at 20% for our application, only at thresholds at or below 0.71, with a corresponding average total precision of 81.2% below this point. In contrast, our CNN-XC models had a total recall of over 20% at all thresholds; and, at the 0.99 threshold, average total precision was 77.1% and average total recall was 28.2% among models (Fig. 5B). Since BirdNET was not trained with data from our domain, these comparisons are not strictly appropriate, yet they provide assurance that our models perform well overall relative to this well-developed standard. Our results also highlight the importance of using BirdNET with a geographic filter for improved precision.

3.4. Optimal model selection for each species

We used soundscape data to select the optimal model with the corresponding threshold that produced the highest F0.5 score. We found that different optimal models were chosen for each species (Figs. 6 & 7). For example, MobileNetv2 was selected for CAVI, ResNet101v2 for WEME, and ResNet50v2 for CAQU (Fig. 6). The MobileNet architecture was the most common optimal model in our soundscape data (17 of 37 species; Fig. 7), with an average F0.5 of 0.80 and average threshold of 0.79 (Table 2).

Based on soundscape data, total precision (pooled across species) with optimal CNN-XC model selection was 84.5%. Species precision ranged from 46.9% (SOSP) to 100.0% (COYE, MOUQ, and WIWA), with average species precision of 85.1 ± 13.2% (Fig. 8A). Total recall (pooled across classes) with optimal models was 60.0% and average species

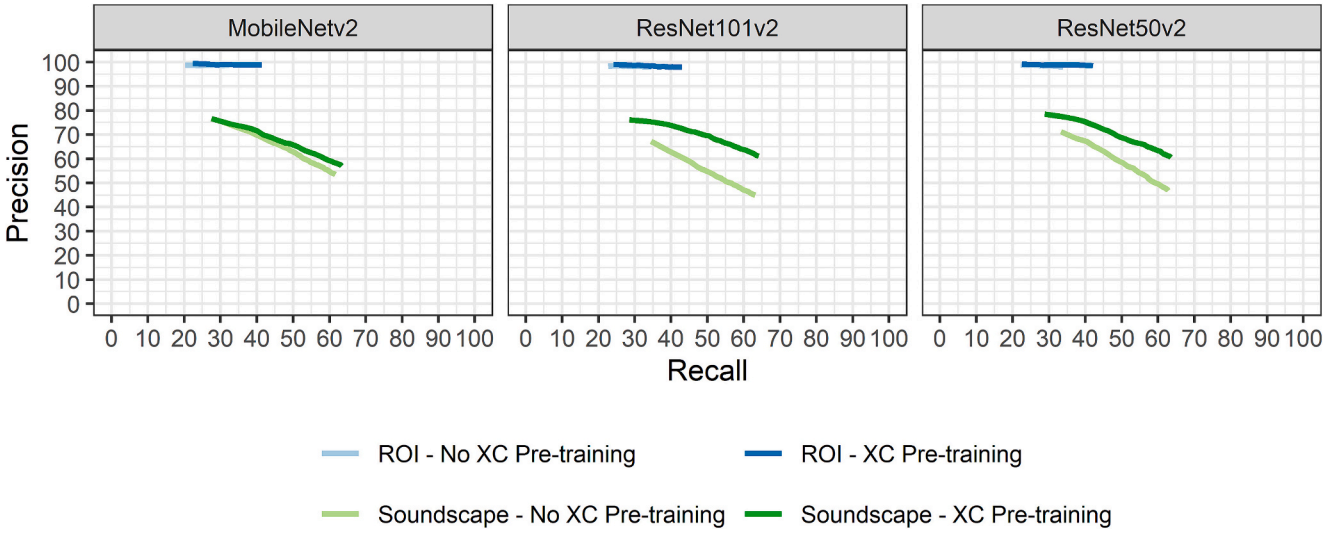


Fig. 3. Total precision-recall curves for each CNN model architecture over a range of probability thresholds (0.65 to 0.99) for XC pre-trained and not pre-trained CNNs, assessed using ROI test and soundscape data across all species.

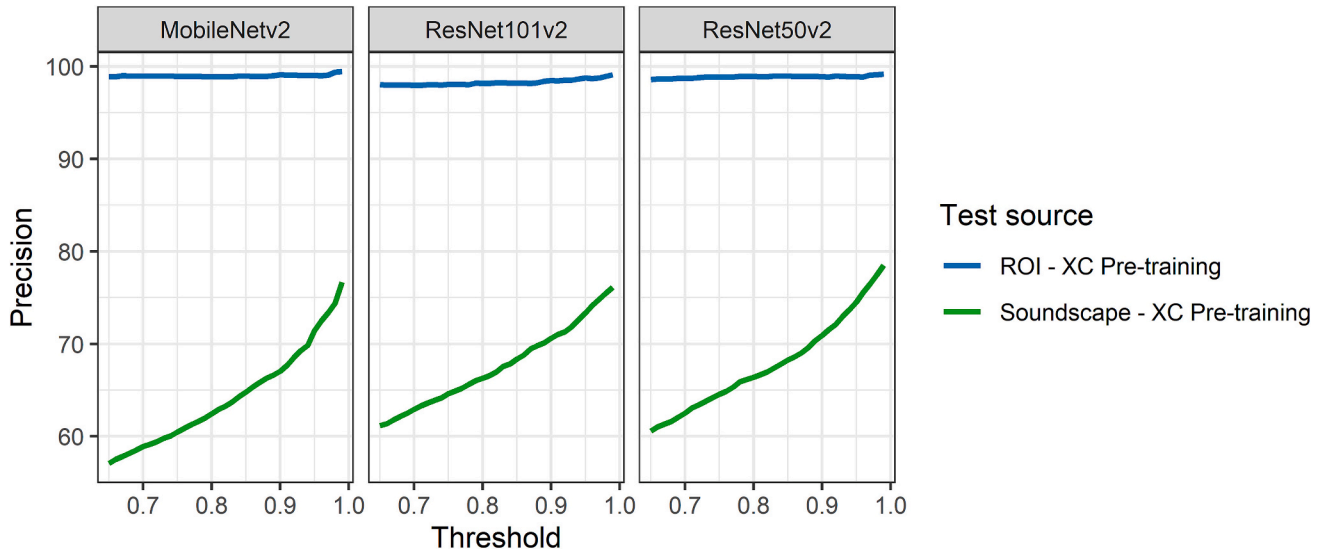


Fig. 4. Total precision over a range of probability thresholds (0.65 to 0.99) for each CNN-XC model architecture with test ROI and soundscape data, respectively, across all species.

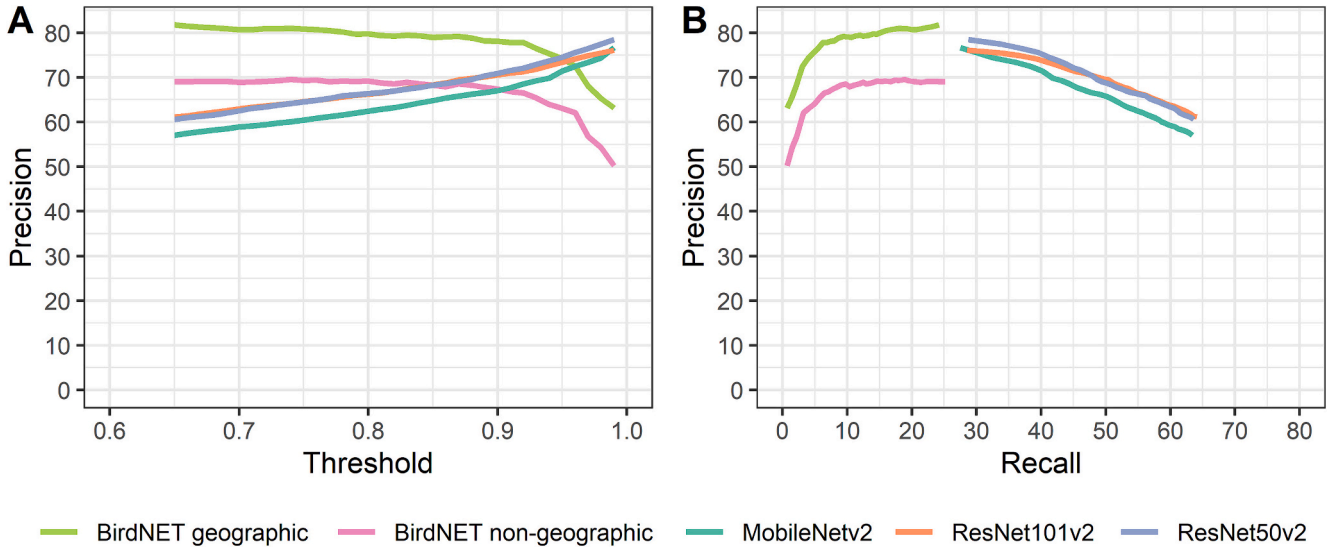


Fig. 5. Accuracy results for all species using soundscape data on study CNN-XC models and BirdNET with and without a geographic filter applied. A) Total precision vs. probability thresholds (0.65 to 0.99). B) Total precision vs. recall.

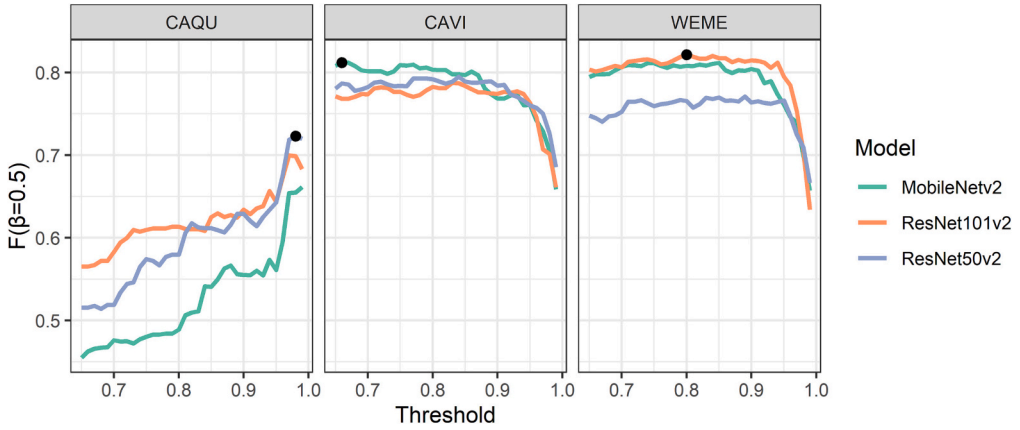


Fig. 6. The variation of F0.5 scores by probability thresholds (0.6500 to 0.9999) for three example species, using soundscape data. The black dot is the optimal model and its respective threshold for a given species.

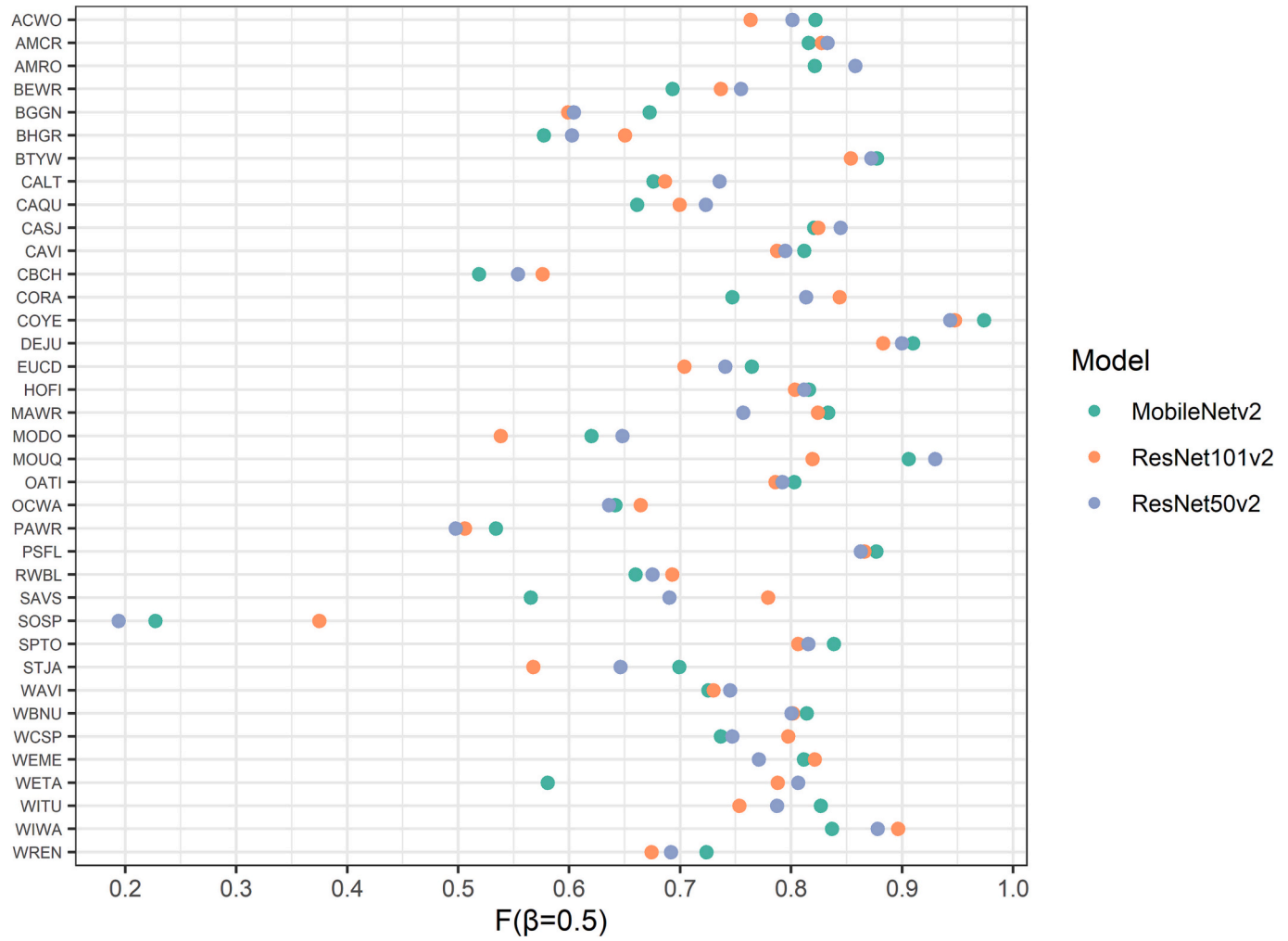


Fig. 7. The variation of F0.5 scores by CNN-XC model with optimal probability thresholds applied to soundscape data (n = 37 species).

Table 2
Average class-level accuracy statistics for optimal species CNN-XC models based on probability thresholds that maximized F0.5 with soundscape data (n = 37 species). Statistics within parentheses are applying the thresholds to ROI data for the same species found in soundscape data.

Model architecture	Number of species	Soundscape threshold	Maximum F0.5	Precision	Recall
MobileNetv2	17	0.79	0.80 (0.69)	86.4% (98.9%)	65.4% (38.4%)
ResNet50v2	9	0.76	0.78 (0.69)	87.5% (99.7%)	56.9% (35.0%)
ResNet101v2	11	0.74	0.72 (0.72)	81.2% (98.3%)	51.6% (40.8%)
All models	37	0.76	0.77 (0.70)	85.1% (98.9%)	59.2% (38.3%)

recall was 59.2 ± 15.0% (Fig. 8B). Species recall ranged from 20.8% (SOSP) to 88.2% (COYE). These levels of recall for each species are acceptable for our biodiversity application that requires accurate species detection at a site level, even if some 2-s detections are missed (i.e., increased number of false negatives that lower recall). With the optimal model selection, overall (pooled across species) F0.5 was 0.78 and F1 was 0.70, while class average F0.5 was 0.77 ± 0.12 and F1 was 0.69 ± 012.

3.5. Influence of soundscape components on bird species detections and model error

With optimal thresholds (Section 3.4) applied to soundscapes, we associated bird detections (TP, FP) and non-detections (FN) with broader sound categories detected by an independent CNN (Quinn et al., 2022) within the same samples of soundscape data (Fig. 9). Detailed logistic regression model results are found in Supplemental Section 3.

Our TP model had highly significant positive effects of Biophony (coeff. = 0.731, z = 9.53, $p < 0.0001$) on correctly identifying bird species, and there was a high prevalence of Biophony associated with TP (Fig. 9). Indeed, the presence of Biophony doubled the odds of a TP detection. This result is expected as Biophony includes birds, as well as other animal vocalizations, and should be trigged by the target species vocalization; however, we are unable to determine from these data if there were overlapping biotic sounds with target bird vocalizations. Other sound components had all highly significant negative effects (all $p < 0.0001$), principally from Geophony (coeff. = −0.57, z = −4.71), followed by (in order), Quiet (coeff. = −0.54, z = −4.84), Interference (coeff. = −0.50, z = −4.79), and Anthropophony (coeff. = − 0.38, z = −4.03). These effects reduced the odds of a TP by about 40% each individually, the worst being Geophony with 43%, and least being Anthrophony with a 31% less chance of a TP detection.

The odds of a FP (vs TP) result were 31.3%. The FP model revealed a highly significant positive influence of Anthropophony (coeff. = 0.89, z = 6.08, $p < 0.0001$), more than doubling the odds (244%) of a FP result

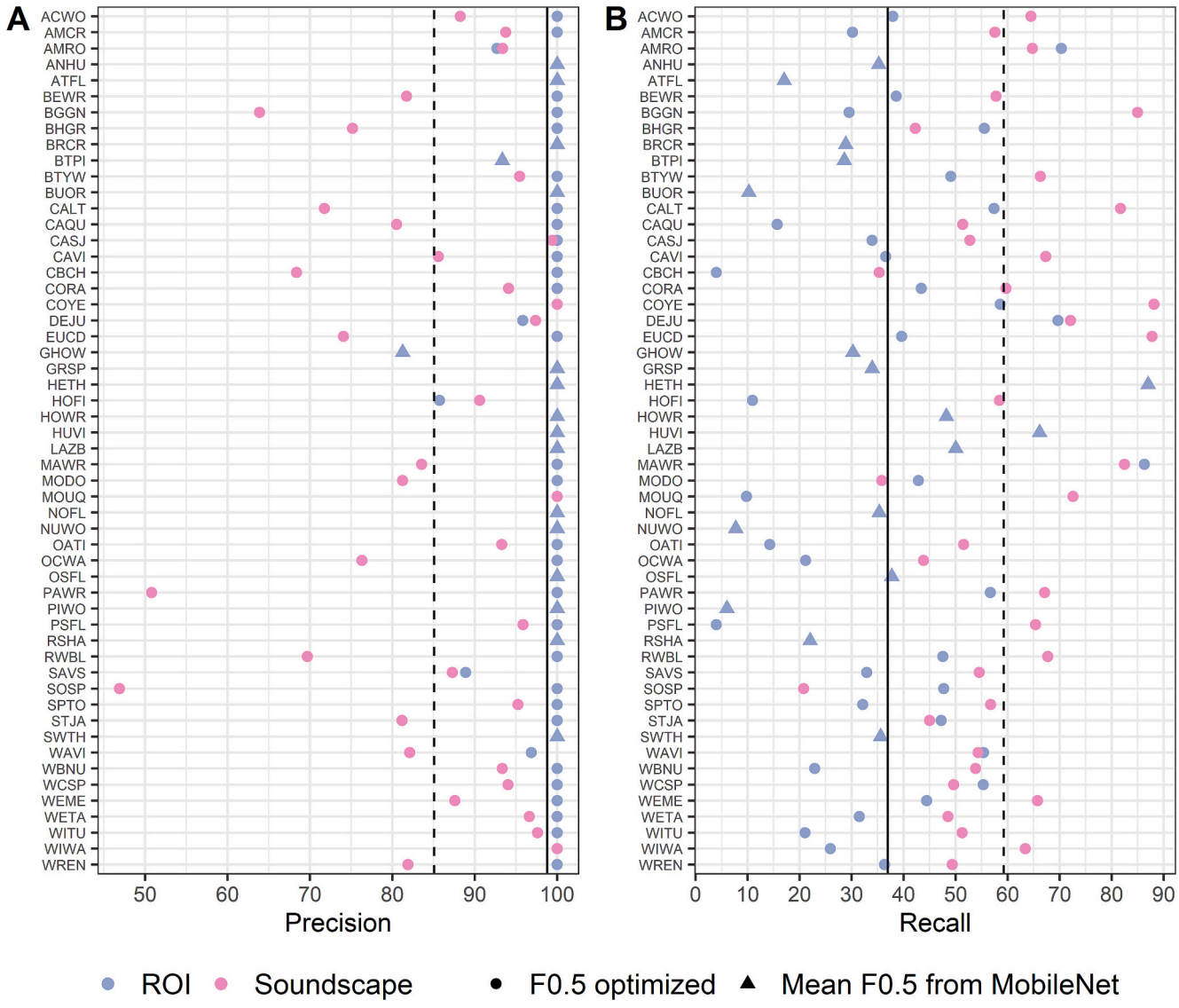


Fig. 8. A) Precision and B) recall for optimal CNN-XC models per species when applied to predictions with ROI test data and soundscapes, respectively. Circles are species with probability thresholds determined by maximum F0.5 in soundscape data ($n = 37$). Triangles are ROI statistics for species that are not in soundscape data ($n = 17$) and use an average soundscape-based optimal threshold with MobileNet (threshold = 0.79). Solid and dashed lines are the average species precision or recall for ROIs ($n = 54$) and Soundscapes ($n = 37$), respectively.

when present. It was followed by (in order) Quiet (coeff. = 0.67, z = 3.87, $p < 0.001$), Geophony (coeff. – 0.58, z = 3.14, $p < 0.01$), and Interference (coeff. = 0.47, z = 2.73, $p < 0.01$). These soundscape components increased the odds of a FP by 95%, 79% and 60%, respectively. Biophony reduced the odds of a FP by 32% (coeff. = 0.25, z = −2.02, $p < 0.05$). Thus, even though there was a high level of Biophony associated with FP (Fig. 9), other sounds (mainly human-generated) and relatively quiet periods were more influential in generating FP in comparison to TP, resulting in lower CNN-XC model precision.

The FN regression model shows that Biophony reduced the odds of a FN result by 61% (coeff. = −0.94, z = −10.84, $p < 0.0001$). Other components significantly increased the odds of a FN result, in order Interference (96% increase in odds, coeff. = 0.67, z = 5.75, $p < 0.001$), Geophony (75% increase, coeff. = 0.56, z = 4.00, $p < 0.001$), and Quiet (40% increase, coeff. = 0.34, z = 2.64, $p < 0.01$). Interference is abrupt high amplitude sounds in our data and significantly increases FN relative to TP, thereby lowering CNN-XC model recall. Anthrophony had no discernible effect on the odds of a FN result.

4. Discussion

4.1. Overall performance of bird vocalization models

An ultimate goal of the *Soundscapes to Landscapes* project is to monitor bird species richness at sample sites using sound recordings from low-cost ARUs. As a step toward this goal, we applied image-based deep learning models (CNNs) for automated detection of 54 bird species from their vocalizations. We prioritized detecting species with confidence at sites, as opposed to capturing all detections in every second of our 1-min recordings. We thus evaluated accuracy with a focus on precision and F0.5 scores, as these measures place more emphasis on true positives and penalize false positives. We evaluated accuracy with test samples, i.e., ROI data, that were focused on individual vocalizations and generated by an automated detection algorithm, and evaluated with consensus-based voting by both expert and regular citizen scientists. Further, we evaluated accuracy against an independent sample of 1-min recordings from the same sound domain with complete labeling of bird calls by our expert citizen scientists, i.e., soundscape data. These

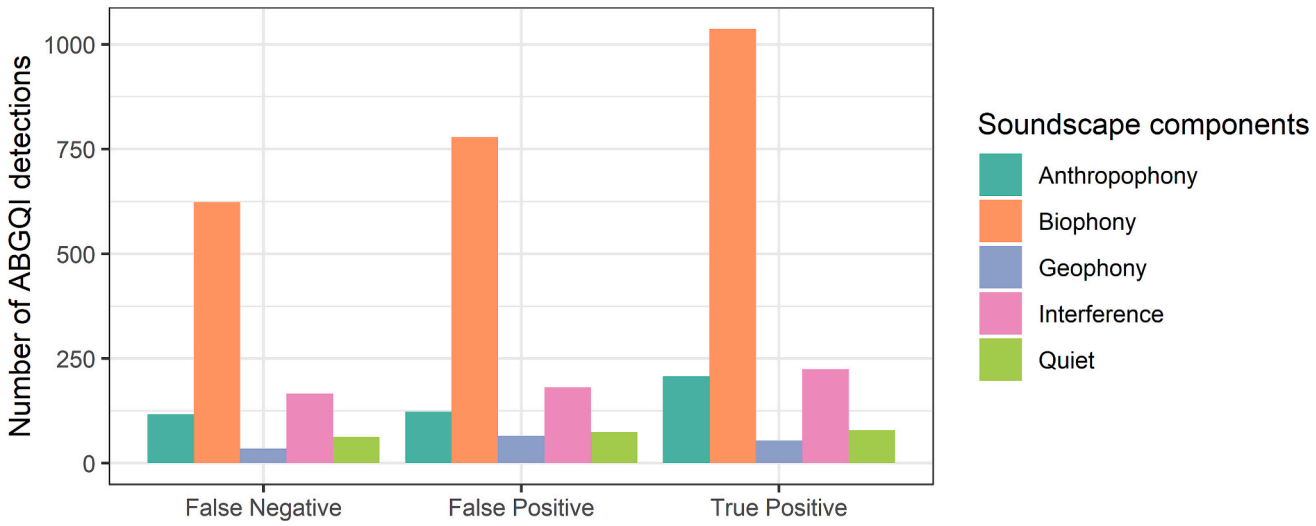


Fig. 9. Distribution of bird detections (TP, FP) and non-detections (FN) with number of soundscape components detected from those samples. Based on optimal thresholds applied to soundscape data (Fig. 8, $n = 37$ species).

data included both non-bird sounds and overlapping vocalizations, thus providing a realistic test of real-world conditions in our study area and environmental conditions, but not outside of this domain (Kahl et al., 2021).

Pre-training with auxiliary XC data resulted in improved evaluation metrics leading to our conclusion that this step helped improve final models. The XC archive is global, open access, and driven by bird enthusiasts, and in the year 2023 included >740,000 recordings and > 10,500 species (xeno-canto.org). These data include recordings with a wide variation in quality and recording devices, which can be beneficial for applied research (Vellinga and Planqué, 2015), and are a significant source of training data for BirdCLEF competitions and BirdNET (Kahl et al., 2020, 2021, 2022). Further, we demonstrated an automated approach to segment sound events in these recordings, thereby making the data accessible for use in CNN development without a high pre-processing cost. A rigorous evaluation of why CNN-XC models increased precision and recall is beyond the scope of our study. However, we speculate that the XC pre-training step helps the CNN model to learn mel-spectrogram image features from bird vocalizations. The fine-tuning step using study-specific mel spectrograms included local properties within our study region (e.g., bird dialects, non-bird sounds, and ARU sound fidelity properties), which improved the final layers of the CNN-XC.

When we selected the best CNN-XC model per species based on the highest F0.5 score, the overall F0.5 score was 0.78 and the class average was 0.77 when tested with soundscape data. In comparison, Kahl et al. (2021) found, using 286 h of fully-annotated audio (i.e., soundscapes), their BirdNET classifier had an F0.5 score of 0.41 on all 24 h and 0.40 on a dawn chorus subset, the latter of which included more and overlapping vocalizations. As with most studies across this research domain, there are differences in testing methodology and data. First, the BirdNET classifier is for 984 bird species across North America and Europe, whereas our study is focused on 54 species in a smaller region. The BirdNET study used randomly selected 1-h recordings for real-world soundscape testing, with predictive averages in 5-s segments. Importantly, these data were from a different domain and included 30 Swift recorders in a one-square-mile area, whereas BirdNET training was based on global recordings from a diverse range of recorders and mostly focused on a single species vocalization. In contrast, our soundscape data were sampled from recordings from 2017 to 2019, from 6 a.m. to 8 p.m. with oversampling of diverse dawn chorus recordings amounting to 628 1-min recordings with 1-s annotations from 343 unique sites around Sonoma County. When we assessed BirdNET directly with our soundscape data, we found the classifier to have much less recall than our CNN models, although BirdNET precision was higher with relatively low levels of recall. This result is expected due to the high domain specificity of CNN models (Pouyanfar et al., 2018). Future research could assess the potential boost in accuracy by using BirdNET fine-tuned with domain-specific sound data.

For comparison with other studies, we also assessed accuracy of our CNN-XC models using independent test data that were a hold-out set from the same data used for training and validation (i.e., ROI data). Our ROI total precision (0.99 threshold) was over 99% and class-average precision was over 97% for all three model architectures. This result is similar to LeBien et al. (2020), who used the same Arbimon pattern-matching routine used in our study and trained a multi-species CNN classifier (24 bird and frog species) that, when tested with ROIs, had a total precision of 98% and class-average precision of 90% at a 0.99 probability threshold. In a study of 14 forest birds and mammals, Ruff et al. (2021) trained a CNN model with precision exceeding 90% at high probability thresholds when tested against spectrograms with up to four target classes including birds, mammals, and noise. The use of a hold-out set for testing is typical in the bioacoustic classification field, yet these tests are positively biased in that sound clips generally come from high-quality recordings and include target species with no overlapping sounds (Kahl et al., 2021). Our use of pattern matching to collect ROIs focused clip selection on a particular frequency bandwidth and duration, as determined by the template vocalization. We scanned 66% of our sound archive with these templates, thereby encompassing high variation in soundscapes from our study area, including non-bird sounds. However, since we selected highly correlated matches, clips generally do not include overlapping bird vocalizations in the frequency band, faint vocalizations, or excessive noise. In contrast, our soundscape data included more variation in test cases, including faint calls, ambient noise, and overlapping vocalizations, generating a more robust accuracy assessment. To our knowledge, no studies have shown this decline in accuracy between hold-out test data (ROI) and soundscapes from the same sound domain. Testing against soundscape data from outside the domain can further yield more insight into the transferability of classifiers to new conditions, as found in recent BirdCLEF competitions (Kahl et al., 2020, 2021, 2022).

While many studies in this domain use curated recordings, often from natural habitats, our sites span a range of land cover and land use types across an urban-wildland gradient, and thus include high levels of non-target biophony (e.g., other birds, insects, amphibians, mammals), anthropophony, geophony, and physical interference noise (Quinn

et al., 2022). Since many soundscapes include these extraneous sounds, particularly anthropophony (Pijanowski et al., 2011), we include this soundscape variation in our analysis to provide an additional real-world assessment of classification accuracy. Our ABGQI CNN analysis showed that non-biotic sounds were significantly associated with increased error in our bird species classification. In particular, the class Anthropophony was associated with false species detections (FP), while the classes Interference and Geophony were associated with missing detections (FN). In our study area, Interference was found in ~10% of all recordings, and was highest between 11 a.m. and 3 p.m. when wind gusts increase and cause recorders and branches to sway, causing physical contact with the recorder or bag (Quinn et al., 2022). This points to the need to secure recorders with straps and place them in locations with no branches, something not always implemented in our citizen science project.

4.2. Mixture of experts (MoE)

In our study, we used three CNN model architectures as a MoE, selecting the best model for a given species based on the highest ROI F0.5 score and coincident probability threshold. We found the MobileNetv2 model architecture was chosen for 46% of species, but with an edge in F0.5 score, yet other species favored ResNet50v2 and ResNet101v2. There are other forms of combining MoE that we did not explore, such as averaging model probabilities (Henkel et al., 2021) or voting (Sampathkumar and Kowerko, 2021) which yielded slight F1 improvements in the BirdCLEF 2021 competition. Other studies in this domain have also found improvements in species classification with ensembles of classifiers. For example, Nanni et al. (2020) found ensembles of CNNs, fine-tuned with bird, bat, or whale vocalizations, outperformed single CNNs. The best ensemble for birds included both automatically learned features from the CNN and hand-crafted features (i.e., acoustic and visual statistics). The ensemble model used spectrograms as input and scattergrams and harmonic and pressure images. Other studies have shown this ensemble approach using multiple CNN classifiers, including alternate time-frequency image representations and hand-crafted features, had better performance over single mel-spectrogram CNNs when classifying bird vocalizations for 14 to 43 species (Xie et al., 2019; Xie and Zhu, 2019). Further, here we used our ABGQI CNN to independently assess the error in our MoE bird species classifier. Future research could explore incorporating ABGQI into an ensemble species classification. For example, these data could be used to refine sigmoid probability thresholds for certain species, such as those found in urban areas that may have vocalizations influenced by human noise (e.g., overlapping sounds or frequency shifts), or Interference or Geophony could be used to filter predictions with high levels of noise.

4.3. Challenges of applied species-level bird classification

The sound data collected as part of this citizen science-based project are challenging in the applied acoustics classification domain. First, our recordings cover a range of landscapes in an urban-wildland gradient, with overlapping biotic sounds, human-generated noise, wind, rain, and physical interference. In addition, our sound recorders have lower fidelity in terms of signal-to-noise and microphone quality than more expensive recorders (Gibb et al., 2019). We recognized this limitation but given our goal to sample large landscapes through field campaigns relying on citizen scientists, we preferred to use a large number of economical recorders that minimized the risk of broken or lost ARUs adversely affecting our operations. We thus sought to analyze real-world conditions in our study area using these ARUs.

There were some errors in our ROI data (e.g., wrong label, false present), as validations were performed by citizen scientists with variable levels of experience. We minimized these errors through consensus-based voting, which kept species present data accurate compared to expert interpretations (Snyder et al., 2022; Table S2). This approach is like other "crowdsource" data collection platforms, such as Zooniverse (Cox et al., 2015). However, we found the Arbimon pattern-matching routine was extremely inefficient at acquiring species present ROI data in the context of our sound data, target species, and available volunteers with expertise to run matches with templates (Snyder et al., 2022). Modern CNN classifiers require large quantities of labeled reference data, and they generally perform best when trained with labeled data from within the study domain, rather than using external data with different sensor characteristics, vocalization dialects, and environmental conditions. Further research is thus needed in automating the reference data collection step, particularly with cloud-based crowdsourcing platforms that would permit applications to scale.

With over a thousand sites sampled over multiple days through five annual campaigns, our applied bird diversity project generated a large amount of sound data to process in the inference step, resulting in >130 million predictions. Our data volume was three-times greater with our MoE approach than if we had just used a single classifier. In applied settings, studies that use ensemble techniques thus also need to consider the additional processing times required with multiple classifiers, and possibly with additional sound representations (e.g., hand-crafted features, time-frequency images) and overlapping windows (Kahl et al., 2021). Although cloud-based processing services could reduce processing times for this volume of data, they are costly.

There is growing interest in applying CNN classifiers for bird vocalization classification/detection in situations at the edge of a data computation network, such as directly on a smartphone, ARU, or Internet of Things device (Chen and Ran, 2019). These devices have memory, processing, battery, and network constraints that may require CNNs with less complexity than currently used in animal vocalization detection research (Disabato et al., 2021). The MobileNetv2 architecture was specifically designed to be deployed on mobile devices, partially by minimizing memory usage during inference (Sandler et al., 2018). We demonstrated that MobileNetv2 was the optimal classifier for the majority of our study species in our soundscape data, and had the second highest class-average precision (86.4%; lowered by PAWR and BGGN) and highest class-average recall (65.4%) and F0.5 (0.80) in comparison to ResNet architectures (Table 2). We note that without selecting the best CNN-XC model per species, thus removing the influence of difficult species on statistics, average maximum F0.5 was the same for all architectures (0.74). Thus, we conclude that MobileNetv2 could be useful for applications requiring smaller CNN architectures.

We left several issues unexplored, yet identified them as areas for future research. First, we do not know how many additional vocalizations would be detected with higher-fidelity sound recordings, although Toenies and Rich (2021) found that the accuracy of California bird species detections with BirdNET was comparable with AudioMoths and more expensive ARUs. Of particular concern in our study are species that had vocalizations in the low-frequency range (e.g., <1000 dB) that has high instrument noise, such as Band-tailed Pigeon (BTPI), Eurasian Collared-Dove (EUCD), Great Horned Owl (GHOW), and Mourning Dove (MODO). Our recordings included many sites and multiple days, although we limited soundscape analysis to 6 a.m. to 8 p.m. These data had faint vocalizations with overlapping soundscape components (Fig. 9). Our ABGQI CNN analysis showed that true detections were negatively related to the class Quiet, while false negatives and false positives significantly increased due to Quiet. Quiet represents periods of ARU self-noise with no other discernible sounds. This class was more common in recordings from LG ARUs, although our experience is that these recorders tended to have less self-noise than AudioMoths (Quinn et al., 2022). Further, we used mel-spectrograms with RGB color, which is a standard approach in analyses like this, and there could be additional experiments with image representation (for example, see Knight et al., 2020b) that may enhance the signal of faint calls and subsequent detection.

Our Biophony class was broad and did not allow us to explore the effect of overlapping animal vocalizations on model error. We used

sigmoid activations (e.g., as opposed to Softmax) primarily because this approach allowed multiple detections from overlapping vocalizations of target species and directly reported the probability of each detection (Ruff et al., 2021). Beyond our training data augmentation within species, we did not explore other pre-processing techniques that may enhance detectability in cases of overlapping sound events. For example, techniques that simulate overlapping vocalizations and other non-biotic sounds (i.e., mixup) have been demonstrated to benefit CNNs in this domain (Kahl et al., 2021). Despite these obstacles that we identified, we had many detections of our target species, and we speculate that these confounding factors do not significantly affect our ability to measure species richness at the site level, which is a primary goal in the *Soundscapes to Landscapes* project.

Another important issue to explore is intraspecific variation in vocalizations. Several of our target species have different calls, and some (e.g., Steller's Jay, STJA) mimic other species or sounds in their environment. In pattern matching, we used templates for two vocalizations per species for 11 of our target species and three vocalizations for STJA (Table S1), which helped capture more instances of these species in ROIs. When tested with soundscape data, the optimal models for species with two to three vocalization templates had a 0.07 lower F0.5 score (non-significant), 4.7% lower precision (non-significant) and a 12.6% lower recall ($p < 0.05$) than for species with a single vocalization template, indicating that more vocalizations for a species may decrease the ability to retrieve detections (i.e., lower recall). In training CNNs and in evaluating their accuracy, we pooled all predictions from a species together without regard to vocalization type, and thus we did not explore how our CNNs perform on different vocalization types. Further, it is not clear if our models could detect a target species based on vocalizations that were not in our templates. We selected templates that were common vocalizations, and pattern matching was done on a large portion of our sound archive; and thus, we speculate that at the site level we captured our target species despite missing predictions based on non-template vocalizations. Lastly, increasing mel-spectrogram length from the 2 s used in our study would potentially provide more information for the models to identify species and reduce the misclassification of similar vocalizations, yet it also could increase intraspecific variation of vocalizations in the models. Further research is needed on the effect of mel-spectrogram size on CNN performance.

5. Conclusions

Our study had a general goal of demonstrating how low-cost ARUs can be implemented effectively to monitor bird diversity through detection of their calls and songs using image-based (2D) CNN classifiers, i.e., deep learning technology. An important part of this research was a large archive of recordings collected by citizen scientists at a county scale through five annual field campaigns. By spanning an urban-wildland gradient, these sound data encompass the complexity of real-world soundscapes, including overlapping mixtures of biophony, anthropophony and geophony, as well as physical interference, and are lower fidelity than recordings from more expensive units. Moreover, we engaged citizen scientists with field-based expertise in bird vocalization as well as those without prior experience to collect a reference dataset of labeled bird vocalizations. These data were critical to training and testing the CNN models to optimize performance in our study domain. Our application is thus a robust test of a scalable citizen science-driven approach to collect soundscape recordings, label vocalization reference data, and apply modern deep learning technology for capturing biologically meaningful species presence data for basic science and conservation efforts.

To this end, we developed a classification pipeline using three CNN architectures (MoE). When tested against a sample of fully-labeled soundscapes, we conclude:

- Pre-training of CNNs with vocalizations from an external sound archive (xeno-canto) improved accuracy after fine-tuning with vocalizations from our study domain.
- Precision was higher when based on test data held-out from labeled data (ROIs) used for training and validation, relative to these measures from more realistic, 1-min soundscapes.
- The MoE approach to classification was important as species had different optimal CNN models and associated probability thresholds, and the choice of model and threshold for a given species was not known *a priori*.
- Classification errors were associated with non-biotic sounds. In particular, human noise was significantly related to more false positives, leading to reduced precision, while physical interference and geophony were related to more false negatives, which reduced recall.

Modern deep learning technology has had a large impact across scientific disciplines, and engaging citizen scientists can accelerate the development process and extend the reach of applications (Lotfian et al., 2021). In biodiversity applications, the pairing of citizen science data collection with CNN classifiers has been used successfully to automate the identification of animals in camera traps (Willi et al., 2019), plant species and traits (Barré et al., 2017; Schiller et al., 2021), and coral reef habitat (Van Den Bergh et al., 2021). Our project successfully demonstrates an approach that leverages citizen science, acoustic data, and deep learning for collecting large quantities of bird diversity data at a regional scale. The methods developed here could be expanded to include other vocalizing animals, implemented at broader spatial scales, or used as part of long-term monitoring at sentinel sites.

Author Contribution Statement

MC, LS and RS coordinated project management, including citizen science-based field and reference sound data collection. WS and DL contributed substantially to reference data collection. SB developed and implemented the CNN models and CQ performed inference. MC and LS designed and implemented the data processing workflow. LS and MC performed accuracy assessments. MC wrote the first draft of the paper. All co-authors commented on and approved the final manuscript. LS, MC, SG and SN obtained funding for and supervised this work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data depository (DOI: 10.5281/zenodo.6512218) https://doi.org/10.5281/zenodo.6512218. Code depository https://github.com/pointblue/Soundscapes2Landscapes

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecoinf.2023.102065.

References

Acevedo, M.A., Corrada-Bravo, C.J., Corrada-Bravo, H., Villanueva-Rivera, L.J., Aide, T. M., 2009 Sep 1. Automated classification of bird and amphibian calls using machine learning: a comparison of methods. Ecol. Inform. 4 (4), 206–214.

Aide, T.M., Corrada-Bravo, C., Campos-Cerqueira, M., Milan, C., Vega, G., Alvarez, R., 2013 Jul 16. Real-time bioacoustics monitoring and automated species identification. PeerJ. 1, e103 https://doi.org/10.7717/peerj.103.

Aide, T.M., Hernández-Serna, A., Campos-Cerqueira, M., Acevedo-Charry, O., Deichmann, J.L., 2017 Nov. Species richness (of insects) drives the use of acoustic space in the tropics. Remote Sens. 9 (11), 1096.

Araya-Salas, M., Smith-Vidaurre, G., 2017 Feb. warbleR: an R package to streamline analysis of animal acoustic signals. Methods Ecol. Evol. 8 (2), 184–191.

Balantic, C.M., Donovan, T.M., 2020 May 3. Statistical learning mitigation of false positives from template-detected data in automated acoustic wildlife monitoring. Bioacoustics. 29 (3), 296–321. https://doi.org/10.1080/09524622.2019.1605309.

Barré, P., Stöver, B.C., Müller, K.F., Steinhage, V., 2017 Jul 1. LeafNet: a computer vision system for automatic plant species identification. Ecol. Inform. 40, 50–56.

Bermúdez-Cuamatzin, E., Ríos-Chelén, A.A., Gil, D., Garcia, C.M., 2011. Experimental evidence for real-time song frequency shift in response to urban noise in a passerine bird. Biol. Lett. 7 (1), 36–38.

Brandes, T.S., 2008 Sep. Automated sound recording and analysis techniques for bird surveys and conservation. Bird Conserv. Int. 18 (S1), S163–S173.

Campos-Cerqueira, M., Aide, T.M., 2016 Nov. Improving distribution data of threatened species by combining acoustic monitoring and occupancy modelling. Methods Ecol. Evol. 7 (11), 1340–1348.

Campos-Cerqueira, M., Aide, T.M., 2017 Nov 15. Lowland extirpation of anuran populations on a tropical mountain. PeerJ. 5, e4059.

Campos-Cerqueira, M., Arendt, W.J., Wunderle Jr., J.M., Aide, T.M., 2017 Dec. Have bird distributions shifted along an elevational gradient on a tropical mountain? Ecol. Evol. 7 (23), 9914–9924.

Catlin-Groves, C.L., 2012 Jan 1. The citizen science landscape: from volunteers to citizen sensors and beyond. Int. J. Zool. 2012.

Chen, J., Ran, X., 2019 Jul 15. Deep learning with edge computing: a review. Proc. IEEE 107 (8), 1655–1674.

Cox, J., Oh, E.Y., Simmons, B., Lintott, C., Masters, K., Greenhill, A., Graham, G., Holmes, K., 2015 May 13. Defining and measuring success in online citizen science: a case study of Zooniverse projects. Comput. Sci. Eng. 17 (4), 28–41.

Darras, K., Batáry, P., Furnas, B., Celis-Murillo, A., Van Wilgenburg, S.L., Mulyani, Y.A., Tscharntke, T., 2018. Comparing the sampling performance of sound recorders versus point counts in bird surveys: A meta-analysis. J. Appl. Ecol. 55 (6), 2575–2586.

Díaz, S., Settele, J., Brondízio, E.S., Ngo, H.T., Agard, J., Arneth, A., Balvanera, P., Brauman, K.A., Butchart, S.H., Chan, K.M., Garibaldi, L.A., 2019 Dec 13. Pervasive human-driven decline of life on Earth points to the need for transformative change. Science. 366 (6471), eaax3100.

Dirzo, R., Young, H.S., Galetti, M., Ceballos, G., Isaac, N.J., Collen, B., 2014 Jul 25. Defaunation in the Anthropocene. Science. 345 (6195), 401–406.

Disabato, S., Canonaco, G., Flikkema, P.G., Roveri, M., Alippi, C., 2021 Aug 23. Birdsong detection at the edge with deep learning. In: 2021 IEEE International Conference on Smart Computing (SMARTCOMP). IEEE, pp. 9–16.

Furnas, B.J., Callas, R.L., 2015 Feb. Using automated recorders and occupancy models to monitor common forest birds across a large geographic region. J. Wildl. Manag. 79 (2), 325–337.

Gibb, R., Browning, E., Glover-Kapfer, P., Jones, K.E., 2019 Feb. Emerging opportunities and challenges for passive acoustics in ecological assessment and monitoring. Methods Ecol. Evol. 10 (2), 169–185. https://doi.org/10.1111/2041-210X.13101.

Gupta, G., Kshirsagar, M., Zhong, M., Gholami, S., Ferres, J.L., 2021 Aug 24. Comparing recurrent convolutional neural networks for large scale bird species classification. Sci. Rep. 11 (1), 1–2.

He, K., Zhang, X., Ren, S., Sun, J., 2016 Oct 8. Identity mappings in deep residual networks. In: European Conference on Computer Vision. Springer, Cham, pp. 630–645.

Henkel, C., Pfeiffer, P., Singer, P., 2021 Jul 16. Recognizing bird species in diverse soundscapes under weak supervision. arXiv preprint arXiv:2107.07728 1–8.

Hill, A.P., Prince, P., Snaddon, J.L., Doncaster, C.P., Rogers, A., 2019 Oct 1. AudioMoth: a low-cost acoustic device for monitoring biodiversity and the environment. HardwareX. 6, e00073 https://doi.org/10.1016/j.ohx.2019.e00073.

Howard, A., Klinck, H., Dane, S., Kahl, S., Denton, T., 2020. Cornell Birdcall Identification, Kaggle. https://kaggle.com/competitions/birdsong-recognition.

Incze, A., Jancsó, H.B., Szilágyi, Z., Farkas, A., Sulyok, C., 2018 Sep 13. Bird sound recognition using a convolutional neural network. In: 2018 IEEE 16th International Symposium on Intelligent Systems and Informatics (SISY). IEEE, pp. 000295–000300.

Jennings, N., Parsons, S., Pocock, M.J., 2008 May. Human vs. machine: identification of bat species from their echolocation calls by humans and by artificial neural networks. Can. J. Zool. 86 (5), 371–377. https://doi.org/10.1139/Z08-009.

Kahl, S., Clapp, M., Hopping, W.A., Goëau, H., Glotin, H., Planqué, R., Vellinga, W.P., Joly, A., 2020 Sep 22. Overview of BirdCLEF 2020: bird sound recognition in complex acoustic environments. In: CLEF 2020-Conference and Labs of the Evaluation Forum, 2696(262).

Kahl, S., Wood, C.M., Eibl, M., Klinck, H., 2021 Mar 1. BirdNET: a deep learning solution for avian diversity monitoring. Ecol. Inform. 61, 101236.

Kahl, S., Navine, A., Denton, T., Klinck, H., Hart, P., Glotin, H., Goëau, H., Vellinga, W.P., Planqué, R., Joly, A., 2022 Sep 5. Overview of BirdCLEF 2022: endangered bird species recognition in soundscape recordings. In: Proceedings of the Working Notes of CLEF 2022-Conference and Labs of the Evaluation Forum.

Khan, A., Sohail, A., Zahoora, U., Qureshi, A.S., 2020. A survey of the recent architectures of deep convolutional neural networks. Artif. Intell. Rev. 53 (8), 5455–5516.

Knight, E.C., Sòlymos, P., Scott, C., Bayne, E.M., 2020 Oct. Validation prediction: a flexible protocol to increase efficiency of automated acoustic processing for wildlife research. Ecol. Appl. 30 (7), e02140 https://doi.org/10.1002/eap.2140.

Knight, E.C., Poo Hernandez, S., Bayne, E.M., Bulitko, V., Tucker, B.V., 2020 May 3. Pre-processing spectrogram parameters improve the accuracy of bioacoustic classification using convolutional neural networks. Bioacoustics. 29 (3), 337–355.

Lasseck, M., 2019. Bird Species Identification in Soundscapes. CLEF (Working Notes, p. 2380.

LeBien, J., Zhong, M., Campos-Cerqueira, M., Velev, J.P., Dodhia, R., Ferres, J.L., Aide, T. M., 2020 Sep 1. A pipeline for identification of bird and frog species in tropical soundscape recordings using a convolutional neural network. Ecol. Inform. 59, 101113 https://doi.org/10.1016/j.ecoinf.2020.101113.

Leclère, D., Obersteiner, M., Barrett, M., Butchart, S.H., Chaudhary, A., De Palma, A., DeClerck, F.A., Di Marco, M., Doelman, J.C., Dürauer, M., Freeman, R., 2020 Sep. Bending the curve of terrestrial biodiversity needs an integrated strategy. Nature. 585 (7826), 551–556.

Lees, A.C., Haskell, L., Allinson, T., Bezeng, S.B., Burfield, I.J., Renjifo, L.M., Rosenberg, K.V., Viswanathan, A., Butchart, S.H.M., 2022. State of the World's birds. Annu. Rev. Environ. Resour. 47, 1.

Lotfian, M., Ingensand, J., Brovelli, M.A., 2021 Jan. The partnership of citizen science and machine learning: benefits, risks, and future challenges for engagement, data collection, and data quality. Sustainability. 13 (14), 8087.

Luther, D.A., Phillips, J., Derryberry, E.P., 2016. Not so sexy in the city: urban birds adjust songs to noise but compromise vocal performance. Behav. Ecol. 27 (1), 332–340.

Mair, L., Ruete, A., 2016 Jan 28. Explaining spatial variation in the recording effort of citizen science data across multiple taxa. PLoS One 11 (1), e0147796. https://doi.org/10.1371/journal.pone.0147796.

Nanni, L., Costa, Y.M., Aguiar, R.L., Mangolin, R.B., Brahnam, S., Silla, C.N., 2020 Dec. Ensemble of convolutional neural networks to improve animal audio classification. EURASIP J. Audio Speech Music Process. 2020, 1–4.

Oliveira, U., Paglia, A.P., Brescovit, A.D., de Carvalho, C.J., Silva, D.P., Rezende, D.T., Leite, F.S., Batista, J.A., Barbosa, J.P., Stehmann, J.R., Ascher, J.S., 2016 Dec. The strong influence of collection bias on biodiversity knowledge shortfalls of B razilian terrestrial biodiversity. Divers. Distrib. 22 (12), 1232–1244. https://doi.org/10.1111/ddi.12489.

Pereira, H.M., Ferrier, S., Walters, M., Geller, G.N., Jongman, R.H., Scholes, R.J., Bruford, M.W., Brummitt, N., Butchart, S.H., Cardoso, A.C., Coops, N.C., 2013 Jan 18. Essential biodiversity variables. Science. 339 (6117), 277–278.

Pijanowski, B.C., Villanueva-Rivera, L.J., Dumyahn, S.L., Farina, A., Krause, B.L., Napoletano, B.M., Gage, S.H., Pieretti, N., 2011 Mar 1. Soundscape ecology: the science of sound in the landscape. BioScience. 61 (3), 203–216.

Pouyanfar, S., Sadiq, S., Yan, Y., Tian, H., Tao, Y., Reyes, M.P., Shyu, M.L., Chen, S.C., Iyengar, S.S., 2018 Sep 18. A survey on deep learning: algorithms, techniques, and applications. ACM Comput. Surv. (CSUR). 51 (5), 1–36. https://doi.org/10.1145/3234150.

Priyadarshani, N., Marsland, S., Castro, I., 2018 May. Automated birdsong recognition in complex acoustic environments: a review. J. Avian Biol. 49 (5), jav–01447.

Pyle, P., DeSante, D.F., 2003. Four-letter and six-letter alpha codes for birds recorded from the American Ornithologist's union check-list area. North American Bird Bander. 28 (2), 64–79.

Quinn, C.A., Burns, P., Gill, G., Baligar, S., Snyder, R.L., Salas, L., Goetz, S.J., Clark, M.L., 2022 May 1. Soundscape classification with convolutional neural networks reveals temporal and geographic patterns in ecoacoustic data. Ecol. Indic. 138, 108831.

Roca, I.T., Desrochers, L., Giacomazzo, M., Bertolo, A., Bolduc, P., Deschesnes, R., Martin, C.A., Rainville, V., Rheault, G., Proulx, R., 2016. Shifting song frequencies in response to anthropogenic noise: a meta-analysis on birds and anurans. Behav. Ecol. 27 (5), 1269–1274.

Rosenberg, K.V., Dokter, A.M., Blancher, P.J., Sauer, J.R., Smith, A.C., Smith, P.A., Stanton, J.C., Panjabi, A., Helft, L., Parr, M., Marra, P.P., 2019 Oct 4. Decline of the North American avifauna. Science. 366 (6461), 120–124.

Royle, J.A., Link, W.A., 2006 Apr. Generalized site occupancy models allowing for false positive and false negative errors. Ecology. 87 (4), 835–841. https://doi.org/10.1890/0012-9658(2006)87[835:gsomaf]2.0.co;2.

Ruff, Z.J., Lesmeister, D.B., Appel, C.L., Sullivan, C.M., 2021 May 1. Workflow and convolutional neural network for automated identification of animal sounds. Ecol. Indic. 124, 107419.

Salamon, J., Bello, J.P., 2017 Jan 23. Deep convolutional neural networks and data augmentation for environmental sound classification. IEEE Signal Process. Lett. 24 (3), 279–283.

Sampathkumar, A., Kowerko, D., 2021 Sep.. TUC media computing at BirdCLEF 2021: noise augmentation strategies in bird sound classification in combination with DenseNets and ResNets. In: CLEF Working Notes 2021, CLEF: Conference and Labs of the Evaluation Forum.

Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.C., 2018. MobileNetV2: inverted residuals and linear bottlenecks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 4510–4520.

Sang, J., Park, S., Lee, J., 2018, September. Convolutional recurrent neural networks for urban sound classification using raw waveforms. In: 2018 26th European Signal Processing Conference (EUSIPCO). IEEE, pp. 2444–2448.

Sauer, J.R., Pardieck, K.L., Ziolkowski Jr., D.J., Smith, A.C., Hudson, M.A., Rodriguez, V., Berlanga, H., Niven, D.K., Link, W.A., 2017 Aug 1. The first 50 years of the North American breeding bird survey. The Condor: Ornithol. Appl. 119 (3), 576–593.

Schiller, C., Schmidtlein, S., Boonman, C., Moreno-Martínez, A., Kattenborn, T., 2021 Aug 12. Deep learning and citizen science enable automated plant trait predictions from photographs. Sci. Rep. 11 (1), 1–2.

Snyder, R.L., Clark, M., Salas, L., Schackwitz, W., Leland, D., Erickson, T., Stephens, T., Tuffli, T., Tuffli, M., Clas, K., 2022 May 27. The soundscapes to landscapes project: development of a bioacoustics-based monitoring workflow with multiple citizen scientist contributions. Citizen Sci. Theory Pract. 7 (1), 24.

Sonoma County Breeding Bird Atlas, 2020. Madrone Audubon Society. https://www.madroneaudubon.org/birds-sonoma-county-breeding-bird-atlas.php (Last accessed, 1/16/2023).

Stowell, D., 2022 Mar 21. Computational bioacoustics with deep learning: a review and roadmap. PeerJ. 10, e13152.

Stowell, D., Wood, M.D., Pamuła, H., Stylianou, Y., Glotin, H., 2019 Mar. Automatic acoustic detection of birds through deep learning: the first bird audio detection challenge. Methods Ecol. Evol. 10 (3), 368–380.

Sullivan, B.L., Aycrigg, J.L., Barry, J.H., Bonney, R.E., Bruns, N., Cooper, C.B., Damoulas, T., Dhondt, A.A., Dietterich, T., Farnsworth, A., Fink, D., 2014. The eBird enterprise: An integrated approach to development and application of citizen science. Biol. Conserv. 169, 31–40.

Toenies, M., Rich, L.N., 2021 Jan 1. Advancing bird survey efforts through novel recorder technology and automated species identification. California Fish Wildlife. 107, 56–70.

Tulloch, A.I., Szabo, J.K., 2012 Dec 1. A behavioural ecology approach to understand volunteer surveying for citizen science datasets. Emu-Austral Ornithol. 112 (4), 313–325. https://doi.org/10.1071/MU12009.

Tulloch, A.I., Possingham, H.P., Joseph, L.N., Szabo, J., Martin, T.G., 2013 Sep 1. Realising the full potential of citizen science monitoring programs. Biol. Conserv. 165, 128–138. https://doi.org/10.1016/j.biocon.2013.05.025.

Van Den Bergh, J., Chirayath, V., Li, A., Torres-Pérez, J.L., Segal-Rozenhaimer, M., 2021 Apr 21. NeMO-net–gamifying 3D labeling of multi-modal reference datasets to support automated marine habitat mapping. Front. Mar. Sci. 8, 347.

Vellinga, W.P., Planqué, R., 2015 Sep. The Xeno-canto collection and its relation to sound recognition and classification. In: CLEF (Working Notes).

Walters, C.L., Freeman, R., Collen, A., Dietz, C., Brock Fenton, M., Jones, G., Obrist, M.K., Puechmaille, S.J., Sattler, T., Siemers, B.M., Parsons, S., 2012 Oct. A continental-scale tool for acoustic identification of European bats. J. Appl. Ecol. 49 (5), 1064–1074.

Weninger, F., Schuller, B., 2011, May. Audio recognition in the wild: Static and dynamic classification on a real-world database of animal vocalizations. In: 2011 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, pp. 337–340.

Willi, M., Pitman, R.T., Cardoso, A.W., Locke, C., Swanson, A., Boyer, A., Veldthuis, M., Fortson, L., 2019 Jan. Identifying animal species in camera trap images using deep learning and citizen science. Methods Ecol. Evol. 10 (1), 80–91.

Xie, J., Zhu, M., 2019 Jul 1. Handcrafted features and late fusion with deep learning for bird sound classification. Ecol. Inform. 52, 74–81.

Xie, J., Hu, K., Zhu, M., Yu, J., Zhu, Q., 2019 Dec 4. Investigation of different CNN-based models for improved bird sound classification. IEEE Access. 7, 175353–175361.

Yosinski, J., Clune, J., Bengio, Y., Lipson, H., 2014 Nov 6. How transferable are features in deep neural networks? arXiv preprint arXiv:1411.1792 1–9.

Zhang, G., 2020 Oct. Spatial and temporal patterns in volunteer data contribution activities: a case study of eBird. ISPRS Int. J. Geo Inf. 9 (10), 597.

Zhang, Z., Xu, S., Zhang, S., Qiao, T., Cao, S., 2021. Attention based convolutional recurrent neural network for environmental sound classification. Neurocomputing 453, 896–903.

Zhong, M., LeBien, J., Campos-Cerqueira, M., Dodhia, R., Ferres, J.L., Velev, J.P., Aide, T. M., 2020 Sep 1. Multispecies bioacoustic classification using transfer learning of deep convolutional neural networks with pseudo-labeling. Appl. Acoust. 166, 107375 https://doi.org/10.1016/j.apacoust.2020.107375.